

Department of Mathematics
Master thesis in Mathematics

Two problems in Shape Optimization

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Introduction

Shape optimization is a classical field of research, which has inspired deep mathematical theories as well as numerous crucial applications in industry and engineering.

A shape optimization problem consists of finding the shape that minimizes a certain cost functional while satisfying some given constraints. We want to find a solution Ω^* of

$$\min_{\Omega \in \mathcal{U}} \mathcal{J}(\Omega),$$

where $\mathcal{U}$ is a class of subsets of $\mathbb{R}^n$, called a set of admissible shapes, and $\mathcal{J}$ is a functional with values in $\mathbb{R}$ that may depend on the solution of a given partial differential equation, varying with the domain.

In this thesis, we treat two different kinds of optimization problems based on the way the solution of the given partial differential equation varies with respect to the domain.

In the first part of the thesis we describe the general theory for shape optimization problems with Dirichlet condition on the free boundary. Given a bounded open set $\Omega \subseteq \mathbb{R}^n$ and an open subset $\omega \subseteq \Omega$ belonging to the set of the possible shapes $\mathcal{U}$, we consider the elliptic equation

$$\begin{cases} -\Delta y = f \text{ in } \omega \\ y|_{\partial\omega} = 0, \end{cases}$$

where f is a fixed function in $H^{-1}(\Omega)$. Our aim is to consider the minimum problem

$$\min_{\omega \in \mathcal{U}} \mathcal{J}(\omega),$$

where the functional $\mathcal{J}$ depends on the solution y_ω of this equation.

We study our problem under rather general assumptions using the theory of Calculus of Variations. First of all we describe various notions of convergence on

domains of $\mathbb{R}^n$ and results on the continuity of the solution, presenting examples and counterexamples on the existence of a minimum. Then, we introduce the technique of relaxation and we write optimality conditions. We will give an example of the relaxation phenomenon, computing an explicit formulation of the relaxed problem and showing that the minimizer is an element of the relaxed set of the admissible shapes and is not a solution to the original problem (i.e. is not a shape).

In the second part of the thesis we consider a functional $\mathcal{J}$ that depends on the solution y_ω of an elliptic equation with source term that varies with the shape:

$$\begin{cases} -\Delta y = \chi_\omega \text{ in } \omega \\ y|_{\partial\Omega} = 0. \end{cases} \tag{1}$$

We approach this problem using the theory of Optimal Control that allows us to write optimality conditions to characterize the minimum. We study a more specific problem taking the set of all possible shapes as

$$\mathcal{U}_L = \{\omega \subset \Omega \ \mid \ |\omega| \leq L|\Omega|\},$$

where L is a fixed real number such that $0 < L < 1$, and the functional $\mathcal{J}$ is defined by

$$\mathcal{J}(\omega) = \frac{1}{2} \left\| y_\omega - y_d \right\|^2_{L^2(\Omega)}, \tag{2}$$

with $y_d \in L^2(\Omega)$ fixed.

We give examples of existence or nonexistence of the minimum to the problem under consideration.

We call y^0 the solution of the PDEs

$$\begin{cases} -\Delta y^0 = 0 \quad in \ \Omega \\ y|_{\partial\Omega} = 0, \end{cases} \tag{3}$$

that is infact $y^0 = 0$. We call y^1 the solution of

$$\begin{cases} -\Delta y^1 = 1 \quad in \ \Omega \\ y|_{\partial\Omega} = 0. \end{cases} \tag{4}$$

We have the following results

Theorem 0.1. *If $y_d \leq 0$ the optimal solution is $\omega = \emptyset$, thus the function $y_\omega = 0$ in Ω.*

Theorem 0.2. *If $y_d \geq y^1$ then there exists a unique optimal solution χ_ω, with $|\omega| = L|\Omega|$.*

Theorem 0.3. *If $y_d \in H^2(\Omega)$ and $-\Delta y_d \leq 0$ or $-\Delta y_d \geq 1$ there exists a unique optimal solution χ_ω.*

When $-\Delta y_d \in (0,1)$ we will give examples of the 'relaxation phenomenon', in which an optimal domain ω may not exist.

Chapter 1

General formulation for Shape Optimization problems with Dirichlet condition on the free boundary

A shape optimization problem is a minimization problem where the unknown variable runs over a class of domains; then every shape optimization problem can be written in the form

$$\min\{F(\Omega) \ \mid \ \Omega \in \mathcal{U}\},$$

where $\mathcal{U}$ is a class of subsets in $\mathbb{R}^n$, called the set of the admissible shapes, and F is a functional defined on $\mathcal{U}$ with values in R, called shape functional.

This is a typical problem of Calculus of Variations and in order to find a solution we have to answer these main questions:

- Does there exist a minimizer?
- Is it unique?
- How can we characterize it?

To approach the problem using the direct method of Calculus of Variations we need to choice a topology in the set of the admissible shapes that guarantees compactness and we need a lower semicontinuous functional in order to achieve the minimum. We can summarize the steps to get to the solution as:

1. check that $m = \inf\{F(\Omega) \ \mid \ \Omega \in \mathcal{U}\}$ is finite;

2. find a minimizing sequence, i.e. a sequence of domains Ω_n in $\mathcal{U}$ such that $\lim_{n\to\infty} F(\Omega_n) = \inf_{\Omega\in\mathcal{U}} F = m$. Such a sequence always exists if $\mathcal{U}$ is not empty, by the definition of the infimum;

3. define a topology in the set of the admissible shapes $\mathcal{U}$ that has a compacteness property, then we have the existence of the limit $\lim_{n\to\infty} \Omega_n = \Omega^*$;

4. have a lower semicontinuous functional, so that $F(\Omega^*) = m$ and Ω^* is the attempted minimum.

We work with functionals that do not depend directly on the domain Ω, but they are of the form $F(\Omega) = J(u_\Omega)$, where u_Ω is the solution of an elliptic partial differential equation set on Ω. So we need a continuous dependence of the solution u_Ω with respect to Ω, to solve this kind of problem we need to add a fifth step that is

5. have the continuity of the solution u_Ω with respect to domain, in order to have that $\lim_{n\to\infty} u_{\Omega_n} = u_{\Omega^*}$ and so have $\lim_{n\to\infty} J(u_{\Omega_n}) = J(u_{\Omega^*})$.

1.1 Topologies on domains of $\mathbb{R}^n$

In this first part of Chapter 1 we define three main topologies on domains in $\mathbb{R}^n$.

Convergence in the sense of characteristic functions

First we want to make a correspondence between a set E in $\mathbb{R}^n$ and the associated characteristic function χ_E defined by $\chi_E(x) = \begin{cases} 1 & \text{if} \quad x \in E \\ 0 & \text{if} \quad x \notin E. \end{cases}$

The characteristic functions belong to $L^\infty(\mathbb{R}^n)$, and if E has finite Lebesgue measure they belong to $L^p(E)$ for every $1 \le p \le \infty$. It holds the following compactness property

Proposition 1.1. *Let E_n be a sequence of measurable sets in $\mathbb{R}^n$. Then a subsequence can be extracted from the sequence χ_{E_n} that converges weak-* in $L^\infty(\mathbb{R}^n)$ to a function $\chi \in L^\infty(\mathbb{R}^n)$. Moreover, the function χ satisfies $0 \le \chi \le 1$ almost everywhere.*

Proof. For every set $E \neq \emptyset$ it holds that $\|\chi_E\|_\infty = 1$, so every characteristic function belong to the closed unit ball of $L^\infty(\mathbb{R}^n)$. For the Banach Alaoglu theorem the unit ball of the dual space of a Banach space is weakly-* compact and if the space is separable it is sequentially compact. We apply this result with the

separable Banach space $L^1(\mathbb{R}^n)$ and his dual space $L^\infty(\mathbb{R}^n)$, so it does exist a subsequence $\chi_{E_{n_k}}$ that converge to a function $\chi \in L^\infty(\mathbb{R}^n)$.

The positivity is preserved, $\chi_{E_n} \geq 0$ for every set E_n then for every function $\varphi \leq 0$ in $L^1(\mathbb{R}^n)$ we have that $\int_{\mathbb{R}^n} \chi_{E_n}\varphi \leq 0$ for every n. Passing to the weak-* limit we have that $\int_{\mathbb{R}^n} \chi_{E_n}\varphi \to \int_{\mathbb{R}^n} \chi\varphi \leq 0$, so we have $\chi \geq 0$. To prove that $\chi \leq 1$, we can reply the same proof to $1 - \chi_{E_n}$. □

So it always exists a weak-* limit of characteristic function that is not in general a characteristic function. But we have the following result which show that the limit is a characteristic function if we have a convergence in the strong sense.

Proposition 1.2. *If $(E_n)_n$ and E are measurable sets in $\mathbb{R}^n$ such that χ_{E_n} weakly-* converges in $L^\infty(\mathbb{R}^n)$ to χ_E, then $\chi_{E_n} \to \chi_E$ in $L^p_{loc}(\mathbb{R}^n)$ for any $p < +\infty$ and almost everywhere.*

Proof. We want to prove that the integral of $|\chi_{E_n} - \chi_E|^p$ in every compact set tend to zero. It suffit to prove that for every ball B_R, the ball centered in 0 with radius R. By assumption we have that

$$\forall \psi \in L^1(\mathbb{R}^n) \qquad \lim_{n\to+\infty} \int_{\mathbb{R}^n} (\chi_{E_n} - \chi_E)\psi dx = 0.$$

If we consider $\psi = \chi_{B_R}\chi_{E^c}$ we have that

$$\lim_{n\to+\infty} \int_{\mathbb{R}^n} (\chi_{E_n} - \chi_E)\chi_{B_R}\chi_{E^c} dx = \lim_{n\to+\infty} \int_{\mathbb{R}^n} \chi_{E_n}\chi_{B_R}\chi_{E^c} dx = \lim_{n\to+\infty} |B_R \cap (E_n \setminus E)| = 0.$$

If we consider $\psi = \chi_{B_R}$, we have that

$$\lim_{n\to+\infty} \int_{\mathbb{R}^n} (\chi_{E_n} - \chi_E)\chi_{B_R} dx = \lim_{n\to+\infty} |B_R \cap (E_n \setminus E)| - |B_R \cap (E \setminus E_n)| = 0,$$

so also $\lim_{n\to+\infty} |B_R \cap (E \setminus E_n)| = 0$. Now we have the conclusion since

$$\int_{B_R} |\chi_{E_n} - \chi_E|^p = |B_R \cap (E_n \setminus E)| - |B_R \cap (E \setminus E_n)|.$$

□

Now we define the convergence in the sense of the characteristic functions by

Definition 1.3. *Let $(E_n)_{n\geq 1}$ and E be measurable sets in $\mathbb{R}^n$. We say that E_n converges in the sense of characteristic functions to E when n goes to ∞ if*

$$\chi_{E_n} \to \chi_E \qquad \text{in } L^p_{loc}(\mathbb{R}^N), \qquad \forall p \in [1, \infty). \tag{1.1}$$

It's important to notice that $\|\chi_{E_n} - \chi_E\|_{L^p} = \|\chi_{E_n} - \chi_E\|_{L^1}^{\frac{1}{p}}$, so it is sufficient to have convergence in L^1_{loc} to have the convergence in every L^p_{loc} and so in the strong sense of the characteristic functions.

For the proposition 1.2. we know that it suffices the weakly-* convergence in $L^\infty(\mathbb{R}^n)$ to a characteristic function to have the convergence in the sense of the characteristic functions.

Convergence in the sense of Hausdorff

We set B a compact set in $\mathbb{R}^N$. First of all, we have to define the Hausdorff distance on the family $\mathcal{K}_B$ of nonempty compact sets included in B. We denote by $d(.,.)$ the euclidean distance in $\mathbb{R}^N$.

Let K_1 and K_2 be in $\mathcal{K}_B$. We set

$$\forall x \in B, \qquad d(x, K_1) := \inf_{y \in K_1} d(x, y),$$

$$\rho(K_1, K_2) := \sup_{x \in K_1} d(x, K_2),$$

$$d^H(K_1, K_2) := \max(\rho(K_1, K_2), \rho(K_2, K_1)).$$

We know that d^H is a distance on $\mathcal{K}_B$, called *Hausdorff distance*, which makes this space compact.

Now we can define the Hausdorff distance also in the family $\mathcal{O}_B$ of the open sets included in B and we can denote this distance by d_H defined by

$$\forall \Omega_1, \Omega_2 \in \mathcal{O}_B, \qquad d_H(\Omega_1, \Omega_2) := d^H(B \setminus \Omega_1, B \setminus \Omega_2).$$

We can see that the distance d_H is independent with respect to B so we have that

Corollary 1.4. *Let Ω_1, Ω_2 be two open bounded sets in $\mathbb{R}^n$. Then*

$$d_H(\Omega_1, \Omega_2) = max\{\rho(\Omega_2 \setminus \Omega_1, \partial\Omega_2), \rho(\Omega_1 \setminus \Omega_2, \partial\Omega_1)\}.$$

We can define the convergence in the sense of Hausdorff for the class $\mathcal{O}_B$ by

Definition 1.5. *Let $(\Omega_n)_{n\geq 1}$ and Ω be open sets included in B. We say that the sequence Ω_n converges in the sense of Hausdorff to Ω if*

$$d_H(\Omega_n, \Omega) \to 0 \qquad when \quad n \to \infty.$$

We will denote this convergence by $\Omega_n \overset{\mathrm{H}}{\to} \Omega$.

The compactness property follows from the theorem

Theorem 1.6. *Let K_n be a sequence of compact sets contained in a fixed compact set B. Then there exists a compact set K contained in B and a subsequence K_{n_k} that converges in the sense of Hausdorff to K as $k \to \infty$.*

The same result lasts for a sequence $(\Omega_n)_n$ of open sets contained in some fixed compact set B. Then the set $\{\Omega \mid \Omega \subseteq B\}$ is compact for the Hausdorff metric d_H.

To demonstrate this result we have to use this proposition

Proposition 1.7. *If K_1 and K_2 are two compact sets, then*

$$d^H(K_1, K_2) = \|d_{K_1} - d_{K_2}\|_{L^\infty(\mathbb{R}^n)}.$$

Therefore,

$$K_n \overset{\text{H}}{\to} K \iff d_{K_n} - d_K \text{ converges uniformly to 0 in } \mathbb{R}^n.$$

where $d_{K_1}(x) = d(x, K_1)$ is the distance from K_1, the same for d_{K_2}.

Proof. (Theorem 1.6) We want to use the proposition above and prove that d_{K_n} converges uniformly on B to d_K. Let us consider the functions d_{K_n} in the space $C(B)$ of the continuous functions on B endowed with the uniform norm. The sequence d_{K_n} is:

1. uniformly bounded by the diameter of B;
2. equicontinuous, $|d_{K_n}(x) - d_{K_n}(y)| \leq d(x, y) \quad \forall n$.

According to the Ascoli-Arzelà theorem there exists a subsequence which converges uniformly on B to a function $f \in C(B)$. We want to say that $f = d_K$ for $K = \{x \in B \mid f(x) = 0\}$ that is compact in B. First of all $|d_{K_n}(x) - d_{K_n}(y)| \leq d(x, y)$, passing to the limit $|d_{K_n}(x) - d_{K_n}(y)| \leq d(x, y)$ so for every $y \in K$ $|f(x)| \leq d(x, y)$. The function f is positive because it is a uniform limit of positive functions, so we have

$$f(x) \leq d_K(y).$$

Now we consider the sequence of points $x_n \in K_n$ such that $d(x, K_n) = d(x, x_n)$, this is a sequence in B which is compact so up to a subsequence it converges to a point $y \in B$. So

$$f(x) = \lim d_{k_{n_k}}(x) = \lim d(x, x_{n_k}) = d(x, y).$$

If $y \in K$ we have the other inequality, but $f(y) = \lim d_{k_{n_k}}(x) \leq d(y, x_{n_k}) = 0$ so $y \in K$ and

$$f(x) \geq d_K(x).$$

□

We recall here some results linked to the Hausdorff convergence that are

Proposition 1.8. *Let Ω_n be a sequence of open sets which converges in the Hausdorff topology to the open set Ω and let $x \in \partial\Omega$. Then there exists a sequence of points x_n, with $x_n \in \partial\Omega_n$, that converges to x.*

Proof. For the purpose of contradiction we have $x \in \partial\Omega$ and we suppose that its distance to $\partial\Omega_n$ does not converge to 0. Then there exists a closed ball $\bar{B}(x, \varepsilon)$ and a subsequence Ω_{n_k} whose intersection is empty. By connectedness $\bar{B}(x, \varepsilon)$ is contained in Ω_{n_k} or in its complement for every k. The Hausdorff convergence is stable for the inclusion so $\bar{B}(x, \varepsilon)$ is contained in Ω or in its complement, which contradicts $x \in \partial\Omega$. □

Proposition 1.9. *If the sequence of open sets Ω_n converges in the Hausdorff topology to the open set Ω and if K is a compact set contained in Ω, then K is contained in Ω_n for n large enough.*

Proof. We have $\inf_{x \in K} d(\cdot, B \setminus \Omega) > 0$ and $d(\cdot, B \setminus \Omega) \leq d(\cdot, B \setminus \Omega_n) + d^H(B \setminus \Omega_n, B \setminus \Omega)$, so $0 < d(\cdot, B \setminus \Omega_n)$ for large n. □

Proposition 1.10. *Let Ω_n and Ω be open sets included in some fixed set B. If $\Omega_n \xrightarrow{\text{H}} \Omega$ then*

(i) $|\Omega \setminus \Omega_n| \to 0$;

(ii) $\chi_\Omega \leq \liminf \chi_{\Omega_n}$ *a.e.* .

Proof. We know that $d^H(B \setminus \Omega_n, B \setminus \Omega) \to 0$ and so do $\rho_n = \rho(B \setminus \Omega_n, B \setminus \Omega) \to 0)$. We fix a sequence of number ε_n grater than ρ_n, then $\chi_{\Omega \setminus \Omega_n} = \chi_{(B \setminus \Omega_n) \setminus (B \setminus \Omega)} \leq \chi_{\{x:\ 0 < d(x, B \setminus \Omega) < \varepsilon_n\}}$. This set converges to the empty set, so for the theorem of monotone convergence we have that

$$\lim_{n \to \infty} \int_B \chi_{\Omega \setminus \Omega_n} = \lim_{n \to \infty} |\Omega \setminus \Omega_n| = 0.$$

For the second point, we have $\chi_\Omega = \chi_{\Omega \setminus \Omega_n} + \chi_{\Omega \cap \Omega_n} \leq \chi_{\Omega \setminus \Omega_n} + \chi_{\Omega_n}$, since the first addend tend to zero we have

$$\chi_\Omega \leq \liminf \chi_{\Omega_n}.$$

□

Convergence in the sense of compact sets

Definition 1.11. *Let $(\Omega_n)_{n\geq 1}$ and Ω be open sets in $\mathbb{R}^N$. We say that Ω_n converges in the sense of compact sets to Ω, $\Omega_n \overset{\text{K}}{\to} \Omega$, if*

1. *for all K compact $\subseteq \Omega$, we have $K \subseteq \Omega_n$ for n large enough;*
2. *for all L compact $\subseteq \bar{\Omega}^c$, we have $L \subseteq \bar{\Omega_n}^c$ for n large enough.*

An important drawback is that the limit is not unique, if a sequence Ω_n converges in the sense of compact sets to Ω then it converges to any open set $A \subset \Omega$ such that $\bar{A} = \bar{\Omega}$.

1.1.1 Perimeter of measurable sets

Another important thing to be defined is the perimeter of a measurable set. Let D be any open set of $\mathbb{R}^N$.

Definition 1.12. *Let Ω be a measurable set in D. The perimeter of Ω relative to D (or simply perimeter if $D = \mathbb{R}^N$), is the number defined by*

$$P_D(\Omega) = \sup\{\int_\Omega div(\varphi)dx; \quad \varphi \in \mathcal{D}(D, \mathbb{R}^N) \quad \text{s.t.} \quad \|\varphi\|_\infty \leq 1\}. \tag{1.2}$$

We can rewrite the integral in this way:

$$\begin{aligned}\int_\Omega div(\varphi)dx &= \int_D \chi_\Omega \left(\sum_{1=1}^n \frac{\partial \varphi_i}{\partial x_i}\right) = \left\langle \chi_\Omega, \sum_{1=1}^n \frac{\partial \varphi_i}{\partial x_i} \right\rangle_{\mathcal{D}'\times\mathcal{D}} \\ &= -\sum_{1=1}^n \left\langle \frac{\partial \chi_\Omega}{\partial x_i}, \varphi_i \right\rangle_{\mathcal{D}'\times\mathcal{D}} = -\langle \nabla \chi_\Omega, \varphi\rangle_{\mathcal{D}'\times\mathcal{D}}. \end{aligned} \tag{1.3}$$

We notice that this definition coincides with the expected usual definition of perimeter if Ω is regular, i.e. a bounded open set of class C^1, and is equal to the measure of the boundary of Ω. So in this case is worth that $P_D(\Omega) = \int_{\partial\Omega\cap D} d\sigma$, where $d\sigma$ is the elementary surface element.

Now we can define the total variation of a Radon finite measure on $\mathbb{R}^n$, i.e. $\mu = (\mu_1, ..., \mu_n) \in \mathcal{D}'(D, \mathbb{R}^n)$, by

$$\|\mu\|_1 = \sup\{\langle \mu, \varphi\rangle_{\mathcal{D}'\times\mathcal{D}}; \quad \varphi \in \mathcal{D}(D, \mathbb{R}^N) \quad \text{s.t.} \quad \|\varphi\|_\infty \leq 1\}. \tag{1.4}$$

We have the following proposition

Proposition 1.13. *Let $\mu \in \mathcal{D}'(D, \mathbb{R}^n)$, then μ is a Radon measure with finite total mass if and only if the variation of μ is finite.*

We will denote by $\mathcal{M}_b(D)$ the space of the real Radon measures on D with finite total mass, and $\mathcal{M}_b(D, \mathbb{R}^n)$. In other words μ is a distribution of order 0, in fact there is a correspondence between the distributions of order 0 in D and the Radon measures on D with finite total mass.

We have from (1.3) that the perimeter $P_D(\Omega)$ is finite if and only if $\nabla\chi_\Omega$ is a measure of finite total mass and we have the relation

$$P_D(\Omega) = \|\nabla\chi_\Omega\|_1 . \tag{1.5}$$

The perimeter is lower semicontinuous (l.s.c.) for the strong convergence of characteristic functions

Proposition 1.14. *Let Ω_n and Ω be measurable and bounded sets in $\mathbb{R}^n$.*

If $\chi_{\Omega_n} \to \chi_\Omega$ in $L^1_{loc}(D)$, then $|\Omega| \leq \liminf |\Omega_n|$ and $P_D(\Omega) \leq \liminf P_D(\Omega_n)$.

We recall that $\mathcal{M}_b(D)$ is the dual space of $C_0(D)$ endowed with the uniform norm, where $C_0(D)$ is the linear space of continuous functions on D that tend to 0 at the boundary of D. This is a separable space, therefore the unit ball of $\mathcal{M}_b(D)$ is sequentially compact for the weak-* topology $\sigma(\mathcal{M}_b(D), C_0(D))$, more precisely if μ_n is a sequence of Radon measures on D such that $\|\mu_n\|_1 \leq C$ there exist a subsequence such that $\mu_{n_k} \overset{*}{\rightharpoonup} \mu$ in the topology $\sigma(\mathcal{M}_b(D), C_0(D))$.

We have this result

Theorem 1.15. *Let Ω_n be a sequence of measurable subsets of an open set D in $\mathbb{R}^n$. Assume that*

$$|\Omega_n| + P_D(\Omega_n) \leq C \qquad \textit{independently of } n.$$

Then there exist a measurable set $\Omega \subseteq D$ and a subsequence Ω_{n_k} such that

$$\chi_{\Omega_{n_k}} \to \chi_\Omega \qquad in \qquad L^1_{loc}(D),$$

$$\nabla\chi_{\Omega_{n_k}} \overset{*}{\rightharpoonup} \chi_\Omega \qquad in \qquad \sigma(\mathcal{M}_b(D)^n, C_0(D)^n).$$

1.1.2 Definition of Capacity

We need to recall some preliminary definitions of the space H^1_0 and H^{-1}. We fix D an open subset of $\mathbb{R}^n$, the Sobolev space $H^1(D) = W^{1,2}(D)$ is

$$H^1(D) := \{u \in L^2(D) \ s.t. \ \nabla u \in L^2(D)^n\},$$

the space of the function u in $L^2(D)$ whose derivative ∇u is to be understood in the sense of distributions is also in $L^2(D)^n$. We equip this space with the norm

$$\|u\|_{H^1}^2 = \|u\|_{L^2} + \|\nabla u\|_{(L^2)^n} = \int_D u^2 + |\nabla u|^2 dx.$$

We denote by $H_0^1(D)$ the closure of C_0^∞ in $H^1(D)$.

The dual space of $H_0^1(D)$ is identified with $H_0^1(D)' = H^{-1}(D)$ where the space H^{-1} is

$$H^{-1}(D) = \{T \in \mathcal{D}'(D) | T = f_0 + \sum_{1 \leq i \leq n} \frac{\partial f_i}{\partial x_i}, f_0, f_i \in L^2(D)\},$$

equipped with the norm

$$\|T\|_{H^{-1}}^2 = \inf\{ \sum_{0 \leq i \leq n} \|f_i\|_{L^2}^2 \}.$$

If D is bounded, from the Poincaré's inequality there is an embedding of $H_0^1(D)$ into $L^2(D)$ which is compact, so for every bounded sequence in $H_0^1(D)$ there exist a subsequence that converges in $L^2(D)$.

The space $H_0^1(D)$ is usually equipped with a norm that is equivalent to the H^1-norm that is

$$\|u\|_{H_0^1} = \left(\int_D |\nabla u|^2 \right)^{\frac{1}{2}},$$

and it is an Hilbert space.

Now we are ready to define the capacity of a set E in $\mathbb{R}^n$

Definition 1.16. *For any $E \subseteq \mathbb{R}^n$,*

$$\text{cap}(E) = \inf\{ \int_D \nu^2 + |\nabla \nu|^2 dx \ \mid \ \nu \in H^1(\mathbb{R}^n), \nu \geq 1 \quad \textit{a.e. in a neighborhood of } E\}.$$

Thus, cap$(E) = 0$ is equivalent to the existence of a sequence ν_n converging to 0 in $H^1(\mathbb{R}^n)$ and greater than or equal to 1 on a neighborhood of E.

When D is a bounded open subset of $\mathbb{R}^n$ we define the "capacity relative to D" associated with the space $H_0^1(D)$ in the same way as we did for the "global" capacity but with the norm $\| \cdot \|_{H_0^1}$.

Definition 1.17. *Given D a bounded set in $\mathbb{R}^n$, for any $E \subseteq D$,*

$$\text{cap}_D(E) = \inf\{ \int_D |\nabla \nu|^2 dx \ \mid \ \nu \in H_0^1(D), \nu \geq 1 \quad \textit{a.e. in a neighborhood of } E\}.$$

In dimension 1 only the empty set is of zero capacity. More in general we have that a d-dimensional manifold, or a subset $E \subseteq \mathbb{R}^n$ contained in a d-dimensional manifold, has zero capacity if and only if $d \leq n-2$.

Here are some definitions:

Definition 1.18. *Given $E \subseteq \mathbb{R}^n$, we will say that a property holds quasi-everywhere (q.e.) in E if it holds everywhere on E except on a subset of zero capacity.*

Definition 1.19. *A subset E of $\mathbb{R}^n$ is said to be quasi-open if for every $\varepsilon > 0$ there exists an open subset $A_\varepsilon \supseteq A$ of $\mathbb{R}^n$ such that $\operatorname{cap}(A_\varepsilon \triangle A) < \varepsilon$, where $\triangle$ denotes the symmetric difference of sets.*

Observation: open sets are obviously quasi-open sets.

Definition 1.20. *A function $f : D \to \mathbb{R}$ is said to be quasi-continuous if for every $\varepsilon > 0$ there exists a continuous function $f_\varepsilon : D \to \mathbb{R}$ such that $\operatorname{cap}(\{f \neq f_\varepsilon\}) < \varepsilon$.*

It's true that every function u of the Sobolev space $H^1(D)$ has a quasi-continuous representative $\tilde{u}$, which is uniquely defined up to a set of capacity zero.

Thanks to the definition of the quasi-open we can characterize the space $H_0^1(D)$ through this theorem

Theorem 1.21. *Let D be an open set of $\mathbb{R}^n$ and Ω a subset of D. A function u in $H^1(D)$ belongs to $H_0^1(\Omega)$ if and only if $\tilde{u} = 0$ quasi-everywhere on Ω^c, where $\tilde{u}$ is the quasi-continuous representative of u.*

Thus we have that

Definition 1.22. *Given $\Omega \subset D$, we define*

$$H_0^1(\Omega) = \{u \in H_0^1(D) : \ \tilde{u} = 0 \text{ q.e. in } D \setminus \Omega\}.$$

Theorem 1.23. *Every strongly convergent sequence in $H^1(\mathbb{R}^n)$ has a subsequence converging q.e. in $\mathbb{R}^n$.*

Finally we give an important property linked with the capacity of a set that is called the (α, r)-property of capacity density, defined by

Definition 1.24. *Let $\alpha, r \in (0, +\infty)$. We will say that an open subset $\Omega \subseteq D$ satisfies the (α, r)-property of capacity density if*

$$\forall x \in \partial\Omega, \qquad \frac{\operatorname{cap}(\Omega^c \cap B(x,r) \cap B(x,2r))}{\operatorname{cap}(B(x,r) \cap B(x,2r))} \geq \alpha > 0.$$

For given $\alpha \in (0,1), r_0 > 0$, we will denote by $\mathcal{O}_{\alpha,r_0}$ the family of open subsets of B satisfying the (α, r)-property of capacity density for all $r \in (0, r_0)$.

1.2 Dirichlet problem for the Laplacian

In this section we briefly describe the existence of the weak solution for the boundary value Dirichlet problem for the Laplacian. We consider an open bounded set Ω in $\mathbb{R}^n$, for every function $f \in H^{-1}(\Omega)$ exists a solution in the sense of distributions for the system

$$\begin{cases} -\Delta u = f & \text{in } \Omega \\ u|_{\partial\Omega} = 0. \end{cases}$$

The following proposition holds

Proposition 1.25. *Let Ω be a bounded open subset of $\mathbb{R}^n$ and let $f \in H^{-1}(\Omega)$. Then there exists a unique solution $u = u_\Omega$ to the problem*

$$u \in H_0^1(\Omega), \qquad \langle f, \nu \rangle_{H^{-1}(\Omega), H_0^1(\Omega)} = \int_\Omega \nabla u \nabla \nu \qquad \forall \nu \in H_0^1(\Omega), \tag{1.6}$$

where the notation $\langle f, . \rangle_{H^{-1}(\Omega), H_0^1(\Omega)}$ means the duality product, i.e. the evaluation of the dual form $f \in H^{-1}(\Omega)$ in the element of $H_0^1(\Omega)$.

Moreover, u is the unique solution of

$$u \in H_0^1(\Omega), \qquad -\Delta u = f \quad in \quad \mathcal{D}'(\Omega), \tag{1.7}$$

and also of the minimization problem $\mathcal{J}(u) = \min\{\mathcal{J}(\nu) \ \mid \ \nu \in H_0^1(\Omega)\}$ where

$$\mathcal{J}(\nu) = \frac{1}{2} \int_\Omega |\nabla \nu|^2 - \langle f, \nu \rangle_{H^{-1}(\Omega), H_0^1(\Omega)}. \tag{1.8}$$

Proof. The space $H_0^1(\Omega)$ is an Hilbert space so for the Riez theorem there exists a unique $u \in H_0^1(\Omega)$ such that for every $\nu \in H_0^1(\Omega)$ the evaluation of f in ν is equal to the scalar product in $H_0^1(\Omega)$ between u and ν (we denote it by $(\cdot, \cdot)_{H_0^1}$) so

$$\langle f, \nu \rangle_{H^{-1}, H_0^1} = \int_\Omega \nabla u \nabla \nu,$$

which solves the first point. If we consider $\nu \in C_0^\infty(\Omega)$, by integration by parts it holds

$$\langle f, \nu \rangle_{H^{-1}, H_0^1} = \int_\Omega \nabla u \nabla \nu = -\int_\Omega \Delta u \ \nu = \langle -\Delta u, \nu \rangle_{\mathcal{D}', C_0^\infty},$$

we conclude for density of $C_0^\infty(\Omega)$ in $H_0^1(\Omega)$ that $f = -\Delta u$ as distribution. For

the third point we consider u the solution of the problem

$$\begin{aligned}\mathcal{J}(u+\nu)-\mathcal{J}(u) &= \frac{1}{2}\int_\Omega |\nabla u+\nabla\nu|^2 - \langle f, u+\nu\rangle - \frac{1}{2}\int_\Omega |\nabla u|^2 - \langle f,u\rangle \\ &= \int_\Omega \nabla u\nabla\nu - \langle f, u+\nu\rangle + \frac{1}{2}\int_\Omega |\nabla\nu|^2 \\ &= \frac{1}{2}\int_\Omega |\nabla\nu|^2,\end{aligned}$$

which is strictly positive if ν is nonzero, so u is the minimum. □

If $f \in H^{-1}(\Omega)$ is a positive function the mapping $\Omega \to u_\Omega$ is nondecreasing with respect to the inclusion, if $\Omega_1 \subset \Omega_2 \Rightarrow u_{\Omega_1} \leq u_{\Omega_2}$. We notice that if Ω is a subset of D, with D an open bounded set in $\mathbb{R}^n$, the space $H_0^1(\Omega)$ is a subspace of $H_0^1(D)$ so

$$\forall \nu \in H_0^1(\Omega) \qquad \langle f,\nu\rangle_{H^{-1}(D),H_0^1(D)} = \langle f,\nu\rangle_{H^{-1}(\Omega),H_0^1(\Omega)},$$

and we can denote it simply with $\langle f,\nu\rangle_{H^{-1},H_0^1}$.

1.2.1 Continuity for the Dirichlet problem

We are given a sequence of open subsets $\Omega_n \subseteq D$ converging in some sense to an open subset Ω. When is it possible to claim that u_{Ω_n} converges in some sense to u_Ω? First of all we notice that the solution u_Ω is bounded

Proposition 1.26. *There exists C depending only on D such that, for all open subsets $\Omega \subseteq D$, the solution of (1.6) satisfies*

$$\|u_\Omega\|_{H_0^1(D)} \leq C\|f\|_{H^{-1}(D)}. \tag{1.9}$$

It follows that corollary

Corollary 1.27. *Let $(\Omega_n)_n$ be any sequence of open subsets of D. Then there exists a subsequence and $u^* \in H_0^1(D)$ such that $u_{\Omega_{n_k}}$ converges weakly in $H_0^1(D)$ to u^* as $k \to +\infty$.*

Proof. The proof is a direct consequence of the uniform bound proved in (1.8) and of the weak compactness of the closed unit ball of the Hilbert space $H_0^1(D)$ that is reflexive. □

We would like to know if this weak limit u^* is of the form u_Ω for a subset $\Omega \subseteq D$. We know that if $\Omega_n \to \Omega$ in some sense such that this property holds:

(A) for every compact set $K \subseteq \Omega$, there exists N_K such that for every $n \geq N_K$ we have $K \subseteq \Omega_n$,

the weak limit u^* satisfies the statement (1.6).

We have the following proposition

Proposition 1.28. *Assume (3.28) holds. Then there exist $u^* \in H_0^1(D)$ and a subsequence $u_{\Omega_{n_k}}$ which converges weakly in $H_0^1(D)$ to u^* satisfying*

$$\langle f, \nu \rangle_{H^{-1}(\Omega) \times H_0^1(\Omega)} = \int_\Omega \nabla u \nabla \nu \qquad \forall \nu \in H_0^1(\Omega).$$

To effectively be the solution u_Ω we have to say that $u \in H_0^1(\Omega)$ and it holds only for certain types of convergence that we will study deeper hereafter.

We have the following result:

Proposition 1.29. *Let Ω_n be a nondecreasing sequence of open subsets of D and $\Omega = \bigcup_n \Omega_n$. Then for all $f \in H^{-1}(D)$, u_{Ω_n} converges to u_Ω.*

Proposition 1.30. *Let Ω_n be a sequence of open sets from the family $\mathcal{O}_{\alpha,r_0}$ which converges in the sense of Hausdorff to an open set Ω. Then for all $f \in H^{-1}(D)$, u_{Ω_n} converges to u_Ω.*

We will give other result concerning the continuity of the solution directly in the solution of the examples in the paragraph below, but first we want to define an important convergence.

γ-convergence

We introduce the notion of γ-convergence, given a bounded set D in $\mathbb{R}^n$

Definition 1.31. *(γ-convergence)*

We say that a sequence of open subsets Ω_n of D γ-converges to the open subset $\Omega \subseteq D$ if for all $f \in H^{-1}(\Omega)$, we have $u_{\Omega_n} \to u_\Omega$ in $H_0^1(\Omega)$.

So this is the exact topology for which continuity holds for the solutions of the Dirichlet problem. If we call $R_\Omega : f \to u_\Omega$ the resolvent operator, we have the γ-convergence is equal to the pointwise convergence of R_{Ω_n} to R_Ω.

So we have that have that Ω_n γ-converges to Ω if $R_{\Omega_n}(f)$ converges to $R_\Omega(f)$ strongly in $H_0^1(D)$ for all $f \in H^{-1}(D)$.

We write two important facts.

Theorem 1.32. *Let Ω_n, Ω be open set in D. For every $f \in H^{-1}(D)$ we have $u_{\Omega_n,f} \to u_{\Omega,f}$ strongly in $H_0^1(D)$ (i.e. Ω_n γ-converges to Ω) if and only if for $f \equiv 1$ we have $u_{\Omega_n,1} \to u_{\Omega,1}$ strongly in $H_0^1(D)$.*

So this show that it is important to know what happens in the particular situation where $f \equiv 1$.

Theorem 1.33. *The sequence Ω_n γ-converges to Ω if and only if R_{Ω_n} converges to R_Ω in the operator norm of $\mathcal{L}(L^2(D))$, that is,*

$$\lim_{n\to+\infty} \sup_{\|f\|_{L^2(D)}\leq 1} \|R_{\Omega_n}(f) - R_\Omega(f)\|_{L^2(D)} = 0. \tag{1.10}$$

1.2.2 Examples of existence and nonexistence of the minimum

ε-cone property

First we define the ε-cone property.

Definition 1.34. *Let y be a point in $\mathbb{R}^n$, ζ a unit vector, and ε a positive real number. Let $C(y,\zeta,\varepsilon)$ be the cone of vertex y (without its vertex), of direction ζ and dimension ε, defined by*

$$C(y,\zeta,\varepsilon) = \{\zeta \in \mathbb{R}^n,\ (z-y,\zeta) \geq \cos(\varepsilon)|z-y| \text{ and } 0 < |z-y| < \varepsilon\}.$$

An open set Ω is said to have the ε-cone property if

$$\forall x \in \partial\Omega, \quad \exists \zeta_x \text{ unit vector such that } \forall y \in \bar{\Omega} \cap B(x,\varepsilon),\ C(y,\zeta_x,\varepsilon) \subseteq \Omega.$$

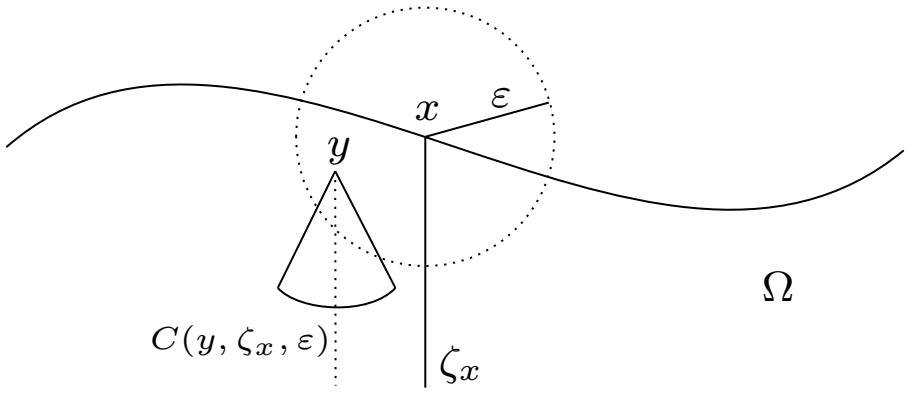


Figure 1.1: The ε-cone property

Proposition 1.35. *A bounded convex open set has the ε-cone property. More generally, this is the case for a bounded open set star shaped with respect to a ball.*

Theorem 1.36. *An open set Ω with a bounded boundary has the ε-cone property if and only if it has a Lipschitz boundary.*

We call $\mathcal{O}_\varepsilon$ the class of all the subsets of D which have the ε-cone property

$$\mathcal{O}_\varepsilon = \{\Omega \subseteq D, \quad \Omega \quad \text{has the} \quad \varepsilon - \textit{cone property}\}.$$

This class is compact with respect to the three main convergence described at the beginning of the chapter:

Theorem 1.37. *Let Ω_n be a sequence of open sets in the class $\mathcal{O}_\varepsilon$. Then there exist an open set $\Omega \in \mathcal{O}_\varepsilon$ and a subsequence Ω_{n_k} that converges to Ω in the sense of Hausdorff, in the sense of characteristic functions and in the sense of compacts.*

With respect to the Hausdorff convergence, the continuity of the solutions holds

Theorem 1.38. *(Chenais)*

Let $(\Omega_n)_n$ be a sequence of open sets of the class $\mathcal{O}_\varepsilon$ defined above converging in the sense of Hausdorff to the open set Ω. Then u_n converges to $u = u_\Omega$.

We consider now a bounded open set $\Omega \in \mathbb{R}^n$ and a subset $\omega \subseteq \Omega$ and the optimization problem $\min_{\omega \in \mathcal{U}_L} \mathcal{J}(\omega)$. We have the functional

$$\mathcal{J}(\omega) = \frac{1}{2} \|y_\omega - y_d\|^2_{L^2(\Omega)},$$

with $y_d \in L^2(\Omega)$ fixed, the set of possible shapes

$$\mathcal{U}_L = \{\omega \subset \Omega \ \mid \ |\omega| \leq L|\Omega| \quad \text{and} \quad \omega \quad \text{has the} \quad \varepsilon - \text{cone property}\}$$

and $y_\omega \in H^1_0(\omega)$ the solution of

$$\begin{cases} -\Delta y = f \quad in\ \Omega \\ y|_{\partial\omega} = 0, \end{cases}$$

with $f \in H^{-1}(\Omega)$.

The functional J is obviously bounded from below from zero. Let ω_n be a minimizing sequence. We use the theorem 1.32. to guarantee the compactness of the set of the admissible shapes in the sense of Hausdorff, in the sense of the characteristic functions and in the sense of the compact sets.

So there is a subset ω such that $\omega_{n_k} \to \omega$. In addition we have that the covergence in the sense of characteristic functions guarantee that the condition $|\omega| \leq L|\Omega|$ is satisfied, because we have $\chi_{\omega_n} \to \chi_\omega$ in $L^1(\bar{\Omega})$ so we have

$$|\omega_{n_k}| = \left|\chi_{\omega_{n_k}}\right|_{L^1(\bar{\Omega})} \to |\chi_\omega|_{L^1(\bar{\Omega})} = |\omega|,$$

every term of the left side is $\leq L|\Omega|$ so for the limit is the same.

We now use the Theorem of Chenais (1.38) to ensure the continuity of the Dirichlet solutions in the case of Hausdorff convergence.

The functional $\mathcal{J}$ is continuous, so l.s.c., therefore is guaranteed that $J(\omega) = \lim J(\omega_{n_k})$. Thus the limit ω is the expected minimum.

Finite number of connected components of the complement

First of all we give a compactness and continuity result for the class of all the subsets of D which have a finite number of connected components of the complement. We call it $\mathcal{O}_M$,

$$\mathcal{O}_M = \{\omega \subset \Omega \ \mid \ \sharp C.C.(\bar{\Omega} \setminus \omega) \leq M\},$$

with M a fixed constant.

We remember the definition of capacity (Definition 1.17) and of the (α, r)-property of capacity density (Definition 1.24).

Only in dimension 2 is true the following property

Proposition 1.39. *Let $K \subseteq \mathbb{R}^2$ be compact, connected, and containing at least 2 points whose distance is equal to a. Then for any $x \in K$,*

$$\forall t \in (0, \frac{a}{2}), \qquad cap(K \cap B(x,t) \cap B(x,2t)) \geq k_2,$$

where k_2 is a "universal" constant.

This is a sequence of the fact that in dimension two any curve has a strictly positive capacity.

Using the compactness property of the class $\mathcal{O}_{\alpha,r}$ for the Hausdorff convergence we can prove that also the set $\mathcal{O}_M = \{\omega \subset \Omega \ \mid \ \sharp C.C.(\bar{\Omega} \setminus \omega) \leq M\}$ is compact respect to this convergence. Now we need the continuity of the Dirichlet solution that follows from this theorem by Šveràk:

Theorem 1.40. *(Šveràk)*

Let Ω_n be a sequence of open sets in the class $\mathcal{O}_M$ which converges in the sense of Hausdorff to Ω. Then for all $f \in H^{-1}(\Omega)$, u_{Ω_n} converges to u_Ω.

We now consider the same functional as above with the set of admissible shapes $\mathcal{O}_M$. Tracing the previous solution we have the existence of a minimun ω. Let ω_n be a minimizing sequence, the compactness property guarantee the existence of a subsequence $\omega_{n_k} \to \omega$. Thanks to the result by Sverak, we have the continuity of the solution only in dimension $d = 2$. The continuity of the functional $\mathcal{J}$ prove that ω is the minimum.

Nonexistence of optimal solutions

In this paragraph we give an explicit example where the existence of an optimal domain does not occur. We consider a bounded open set $\Omega \in \mathbb{R}^n$ and a subset $\omega \subseteq \Omega$. We have the functional

$$\mathcal{J}(\omega) = \int_\Omega |y_\omega - c|^2 dx,$$

with c is a fixed constant, the set of possible shapes

$$\mathcal{A} = \{\omega \subset \Omega \ \mid \ \omega \text{ open}\},$$

and $y_\omega \in H_0^1(\omega)$ the solution of

$$\begin{cases} -\Delta y = 1 \quad in\ \omega \\ y|_{\partial\omega} = 0, \end{cases}$$

with $f \equiv 1 \in H^{-1}(\Omega)$. The optimization problem is

$$\min_{\omega \in \mathcal{A}} \mathcal{J}(\omega). \tag{1.11}$$

We will prove that for small values of the constant c no regular domain ω can solve problem above; We won't give proof of nonexistence of any domain because is slightly more technical and delicate.

We have the following theorem

Theorem 1.41. *If c is small enough, then no smooth domain $\omega \subseteq \Omega$ can solve the optimization problem* (1.11).

Proof. Assume by contradiction that a regular domain ω solves the given optimization problem. Let us also assume first that ω does not coincide with the whole set Ω, so that we can take a point x_0 in Ω which does not belong to the closure $\overline{\omega}$ and

a small ball B_ε of radius ε, centered at x_0 and disjoint form ω. If y_ω denotes the solution of

$$\begin{cases} -\Delta y = 1 & in\ \omega \\ y \in H_0^1(\omega), \end{cases} \tag{1.12}$$

then the solution $y_{\omega \cup B_\varepsilon}$, corresponding to the admissible choice $\omega \cup B_\varepsilon$, is

$$y_{\omega \cup B_\varepsilon}(x) = \begin{cases} y_\omega & if\ x \in \omega, \\ \dfrac{\varepsilon^2 - |x - x_0|^2}{2n} & if\ x \in B_\varepsilon, \\ 0 & otherwise. \end{cases} \tag{1.13}$$

$$\mathcal{J}(y_\omega) = \int_\omega |y_\omega - c|^2 dx + \int_{B_\varepsilon} c^2 dx + \int_{\Omega \setminus (\omega \cup B_\varepsilon)} c^2 dx, \tag{1.14}$$

$$\mathcal{J}(y_{\omega \cup B_\varepsilon}) = \int_\omega |y_\omega - c|^2 dx + \int_{B_\varepsilon} |\frac{\varepsilon^2 - |x - x_0|^2}{2n} - c|^2 dx + \int_{\Omega \setminus (\omega \cup B_\varepsilon)} c^2 dx, \tag{1.15}$$

Comparing the cost $\mathcal{J}(y_\omega)$ to the cost $\mathcal{J}(y_\omega \cup B_\varepsilon)$ and using the minimality of ω then gives

$$\begin{aligned} c^2 |B_\varepsilon| &\leq \int_{B_\varepsilon} |\frac{\varepsilon^2 - |x - x_0|^2}{2n} - c|^2 dx \\ &= n\varepsilon^{-n} |B_\varepsilon| \int_0^\varepsilon |\frac{\varepsilon^2 - r^2}{2n} - c|^2 r^{n-1} dr \\ &= c^2 |B_\varepsilon| + \frac{|B_1|}{4n} \int_0^\varepsilon (\varepsilon^2 - r^2)(\varepsilon^2 - r^2 - 4nc) r^{n-1} dr \end{aligned}$$

which, for a fixed $c > 0$, turns out to be false if ε is small enough. Thus all smooth domains $\omega \subset \Omega$ are ruled out by the argument above. We can now exclude also the case $\omega = \Omega$ if c is small, by comparing for instance the full domain Ω to the empty set. This gives, taking into account that $y_\emptyset = 0$,

$$\mathcal{J}(y_\Omega) = \int_\Omega |y_\Omega - c|^2 dx,$$

$$\mathcal{J}(y_\emptyset) = \int_\Omega c^2 dx,$$

so that we have $\mathcal{J}(y_\emptyset) < \mathcal{J}(y_\Omega)$ if c is sufficiently small. Hence all smooth subdomains of Ω are excluded, and the proof is complete. □

We can minimize the problem also over the set

$$\mathcal{A} = \{\omega \subset \Omega \ \mid \ \omega \text{ quasi-open}\},$$

and $y_\omega \in H_0^1(\omega)$ that has to be intent with the characterization of Definition 1.22. In particular the previous set with the ω open is contained in this second set because an open set is also quasi-open. The formulation of the Dirichlet problem is the same of Proposition 1.24. also in the case of quasi-open sets. We still have no solution for the problem.

We will see in Chapter 2 the relaxed formulation of the problem and we will show that the minimizer is a measure on the relaxed set of the admissible shapes thus not indeed a shape and not a solution of the original problem.

Monotonicity of the functional

We are going to give a general existence result for an optimal shape for functionals that are nonincreasing with respect to inclusion:

$$\mathcal{J}(\Omega_1) \geq \mathcal{J}(\Omega_2) \quad \text{when} \quad \Omega_1 \subset \Omega_2 \subseteq D \tag{1.16}$$

where D is a fixed bounded open set in $\mathbb{R}^n$.

We are going to work with the following class:

$$\mathcal{A}(D) = \{\Omega \subseteq D \ \mid \ \Omega \textit{ quasi-open}\}. \tag{1.17}$$

From Definition 1.22. we can characterize the space $H_0^1(\Omega)$ for $\Omega \subset D$ quasi-open, as

$$H_0^1(\Omega) = \{u \in H_0^1(D) : \ \tilde{u} = 0 \textit{ q.e. in } D \setminus \Omega\}.$$

Then we call w_Ω the solution of the Dirichlet problem

$$\begin{cases} -\Delta w = 1 \textit{ in } \Omega, \\ w \in H_0^1(\Omega), \end{cases} \tag{1.18}$$

so $w_\Omega = R_\Omega(1)$.

We give the following result

Theorem 1.42. *Let $\mathcal{J} : \mathcal{A}(D) \to (-\infty, +\infty]$ be a functional satisfying*

i) $\mathcal{J}$ is lower semicontinuous in the topology of γ-convergence;

ii) $\mathcal{J}$ is nonincreasing for the inclusion: if $\Omega_1 \subset \Omega_2$ then $(\Omega_1) \geq J(\Omega_2)$.

Then for all $c \in (0, |D|)$, the problem

$$\min_{\mathcal{A}_c} \mathcal{J}(\Omega),$$

with

$$\mathcal{A}_C = \{\Omega \in \mathcal{A}(D),\ |\Omega| = c\},$$

has a solution.

To give a proof of the following result we give a new definition and some preliminary propositions.

We now introduce the notion of *weak γ-convergence*

Definition 1.43. *(weak γ-convergence)*

We will say that a sequence of quasi-open sets Ω_n of $\mathcal{A}(D)$ converges weakly to a quasi-open set Ω if the sequence of functions w_{Ω_n} converges weakly in $H_0^1(D)$ to some function w such that $\Omega = \{x :\ w(x) > 0\}$.

In general the limit w of the weak γ-convergence does not coincide with w_Ω. This fact is true only when $\Omega_n \xrightarrow{\gamma} \Omega$.

We have that γ-convergence implies weak γ-convergence.

The set $\mathcal{A}(D)$ is not compact in the topology of γ-convergence, as we will show in chapter 2, but it is sequentially compact for the weak γ-convergence.

Proposition 1.44. *$\mathcal{A}(D)$ is sequentially compact for the weak γ-convergence.*

Proof. To show this we consider a sequence Ω_n of quasi-open sets in $\mathcal{A}(D)$, from Proposition 1.26 we have that w_{Ω_n} is bounded in $H_0^1(D)$. Thus there exists a subsequence converging to some function w. Setting $\Omega = \{x :\ w(x) > 0\}$, we then obtain that the sequence Ω_n weakly γ-converges to Ω. □

We have the two following lemma

Lemma 1.45. *Assume w_{Ω_n} converges weakly in $H_0^1(D)$ to w. Let $v_n \in H_0^1(\Omega_n)$ converge weakly in $H_0^1(D)$ to v. Then $v \in H_0^1(\Omega_w)$*

Lemma 1.46. *Let Ω_n be a sequence of quasi-open subsets of D such that w_{Ω_n} converges weakly in $H_0^1(D)$ to $w \in H_0^1(\Omega)$, where is a quasi-open subset of D. Then there exist a sequence of integers n_k and a sequence of quasi-open sets C_k that γ-converges to Ω with $\Omega_{n_k} \subset C_k \subset D$.*

We will use them to prove the following proposition that is the key point in the proof of Theorem 1.42.

Proposition 1.47. *Let $\mathcal{J} : \mathcal{A}(D) \to (-\infty, +\infty]$ be a functional that is nonincreasing for the inclusion. Then $\mathcal{J}$ is lower semicontinuous in the topology of γ-convergence if and only if it is lower semicontinuous in the topology of weak γ-convergence.*

Proof. Since γ-convergence implies weak γ-convergence it is sufficient to prove that lower semicontinuity in the topology of γ-convergence implies lower semicontinuity in the topology of weak γ-convergence. Thus let Ω_n be a sequence of quasi-open subsets of D that γ-converges weakly to the quasi-open set Ω. Up to a subsequence, we may assume that $L = \lim \mathcal{J}(\Omega_n)$. By definition, w_{Ω_n} converges weakly to $w \in H_0^1(D)$ with $\Omega = \{x : \ w(x) > 0\}$, so that $w \in H_0^1(D)$.

Let n_k and C_k be the sequence associated to the sequence Ω_n according to Lemma 1.46. By monotonicity of $\mathcal{J}$ and thanks to its lower semicontinuity, we have

$$\mathcal{J}(\Omega) \leq \liminf \mathcal{J}(C_k) \leq \liminf \mathcal{J}(\Omega_{n_k}).$$

□

Hence we are now able to demonstrate Theorem 1.42.

Proof. (Theorem 1.42)

Let Ω_n be a minimizing sequence. It weakly γ-converges, up to a subsequence, to a quasi-open set Ω for Proposition 1.44..The functional is lower semi-continuous for the γ-convergence, thus for Proposition 1.47 also for weak γ-convergence. We have that

$$|\Omega| \leq c, \qquad \mathcal{J}(\Omega) \leq \inf_{\mathcal{A}_c} \mathcal{J}(\Omega).$$

If $|\Omega| = c$ then it is the solution. If $|\Omega| < c$, since $\mathcal{J}$ is nonincreasing, any quasi-open subset of D containing Ω and of measure c is a solution. □

We can have the same result also for a functional $\mathcal{J}$ of the form

$$\mathcal{J} = \int_D g(x, u_\Omega) dx,$$

with $u_\Omega = R_\Omega(f)$ the solution of the Dirichlet problem with a function $f \in H_0^1(D)$ s.t. $f \geq 0$ and with g such that

i) $g(x, \cdot)$ is lower semi-continuous in $\Omega \times \mathbb{R}$,

ii) g is nonincreasing,

iii) g satisfies $g(x, s) \geq -a(x) - bs^2$ for suitable $a \in L^1(D)$, $b \in \mathbb{R}$.

We can apply Theorem 1.42. because the hypothesis on g guarantee the lower-semi continuity with respect to the γ-convergence. Since $f \geq 0$ the map $\Omega \to u_\Omega$ is nondecreasing and g is nonincreasing thus the functional $\mathcal{J}$ is nonincreasing too.

We can generalize the atteined result for optimal partition problems of the form

$$\min\{\mathcal{J}(\Omega_1, ..., \Omega_k) \ \mid \ \Omega_i \in \mathcal{A}(D), \ \Omega_i \cap \Omega_j = \emptyset \text{ for } i \neq j\}, \tag{1.19}$$

where k is a fixed positive integer, $\mathcal{A}(D)$ is the class of all admissible domains defined by (1.17), and $\mathcal{J} : \mathcal{A}(D)^k \to [0, +\infty]$ is the cost functional.

We say that $\mathcal{J}$ is monotonically decreasing in the sense of the set inclusion if for all $(\Omega_1, ..., \Omega_k), (\Lambda_1, ..., \Lambda_k) \in \mathcal{A}(D)$ such that $\Omega_i \subset \Lambda_i$ for $i = 1, ..., k$, k in the sense of capacity, i.e. $\text{cap}(\Lambda_i \setminus \Omega_i) = 0$, then

$$\mathcal{J}(\Lambda_1, ..., \Lambda_k) \leq \mathcal{J}(\Omega_1, ..., \Omega_k),$$

where $\Omega_i \subset \Lambda_i$ is in the sense of capacity, i.e. $\text{cap}(\Omega_i \setminus \Lambda_i) = 0$.

We have the same results as above

Theorem 1.48. *Let $\mathcal{J} : \mathcal{A}(D)^k \to [0, +\infty]$ be a shape functional which is monotonically decreasing. Then $\mathcal{J}$ is γ-lower semicontinuous if and only if it is weakly γ-lower semicontinuous.*

Theorem 1.49. *Let $\mathcal{J} : \mathcal{A}(D)^k \to [0, +\infty]$ be a γ-lower semicontinuous shape functional. Then the following optimization problem admits a solution*

$$\min\{\mathcal{J}(\Omega_1, ..., \Omega_k) \ \mid \ \Omega_i \in \mathcal{A}(D), \ \text{cap}(\Omega_i \cap \Omega_j) = 0 \text{ for } i \neq j\}.$$

Hence we conclude that

Theorem 1.50. *If $\mathcal{J} : \mathcal{A}(D)^k \to [0, +\infty]$ is monotonically decreasing and γ-lower semicontinuous, then problem (1.19) admits a solution.*

Chapter 2

Relaxation and optimality condition

In this chapter our aim is to explain the technique of relaxation. If we consider a general optimization problem, it is often very difficult or impossible to find a solution. We want to extend the notion of solution considering a wider class of objects in which we try to minimize the functional, this is the principle of relaxation. This allows us to find an optimal object that replaces an optimal shape when the latter does not exist. The goal then become to write the optimality conditions and we are sometimes able to prove that the optimal relaxed object is indeed a shape.

Before introducing relaxation, let us introduce the notions of γ-convergence, Γ-convergence and Mosco-convergence and let us make the link between these different notions of convergence.

2.1 Different notions of convergence

We recall here the definition of γ-convergence, given a bounded set D in $\mathbb{R}^n$

Definition 2.1. *(γ-convergence)*

We say that a sequence of open subsets Ω_n of D γ-converges to the open subset $\Omega \subseteq D$ if for all $f \in H^{-1}(\Omega)$, we have $u_{\Omega_n} \to u_\Omega$ in $H_0^1(\Omega)$.

And we recall the linked result of Theorem 1.31. and Theorem 1.32. .

Now we define the Mosco-convergence that holds in general for every sequence A_n of closed convex sets of a normed space X, in this case we see the spaces $H_0^1(\Omega_n)$

and $H_0^1(\Omega)$ as convex and closed subset of the Hilbert space $H_0^1(D)$.

Definition 2.2. *(Mosco-convergence)*

The sequence of spaces $H_0^1(\Omega_n)$ converges in the sense of Mosco to the space $H_0^1(\Omega)$ if the following conditions are satisfied:

- *For all $\phi \in H_0^1(\Omega)$ there exists a sequence $\phi_n \in H_0^1(\Omega_n)$ such that ϕ_n converges strongly in $H_0^1(D)$ to ϕ.*
- *For every sequence $\phi_{n_k} \in H_0^1(\Omega_{n_k})$ weakly convergent in $H_0^1(D)$ to a function ϕ we have $\phi \in H_0^1(\Omega)$*

It's important to notice that these two convergences coincide

Proposition 2.3. *Let $(\Omega_n)_n$ be a sequence of open subsets of D. Then Ω_n γ-converges to Ω if and only if $H_0^1(\Omega_n)$ Mosco-converges to $H_0^1(\Omega)$.*

The notion of Γ-convergence instead is a convergence of functionals defined on a metric space X. We ask ourselves what is the right topology on the metric space X in order that the convergence of functionals J_n to J implies the convergence of the minima of J_n to the minima of J.

Definition 2.4. *(Γ-convergence)*

Let (J_n) be a sequence of functionals from a metric space X to $\mathbb{R}$. We say that J_n Γ-converges to a functional J if for any $x \in X$ we have:

(i) $\forall\ x_n \to x \qquad J(x) \leq \liminf_{n\to+\infty} J_n(x_n)$;

(ii) $\exists\ x_n \to x \qquad J(x) \geq \limsup_{n\to+\infty} J_n(x_n)$.

We have an important theorem by Ennio de Giorgi

Theorem 2.5. *Let J_n be a sequence of functionals that Γ-converges to a functional J and that is bounded from below independently of n. Assume there exists a sequence ε_n of positive real numbers converging to 0, and a sequence x_n relatively compact, such that*

$$J_n(x_n) \leq \inf J_n + \varepsilon_n.$$

Then

(i) J has a minimum and $\min J = \lim_{n\to+\infty} \inf J_n$;

(ii) any closure point of the sequence x_n achieves the minimum of J.

We can restate this theorem doing a more generic assumption, it suffit to take a sequence J_n of functional that Γ-converges to J and that is equi-coercive on X,

this means that for every $t \in \mathbb{R}$ there exists a compact set $K_t \subseteq X$ such that

$$\{J_n \leq t\} \subseteq K_t \qquad \forall n.$$

Then G is coercive too and it attains its minimum on X, so we have $\min J = \lim_{n\to+\infty} \inf J_n$ (i.e. point (i) of the theorem above).

Moreover if $x_n \in \text{argmin} J_n$ is a sequence with $x_n \to x$ in X, then $x \in \text{argmin} J$, and if J is not identically $+\infty$ there exists a subsequence of $(x_n) \in \text{argmin} J_n$ which converges to an element of $\text{argmin} J$.

It is interesting to notice that a sequence (J_n) of functionals is equi-coercive in X if and only if there exists a lower semicontinuous coercive function $\Psi : X \to \mathbb{R}$ such that $J_n \geq \Psi$ for all $n \in \mathbb{N}$. Now we consider the subclass of all lower semi-continuous functionals on X that are bounded from below by a coercive function, we call it

$$S_\Psi(X) = \{J : X \to \mathbb{R} \ \mid \ J \text{ is l.s.c., } J \geq \Psi\}.$$

Then the Γ-convergence on $S_\Psi(X)$ is metrizable.

Another important result is the compactness of Γ-convergence:

Theorem 2.6. *If X is separable, then Γ-convergence is sequentially compact: from any sequence of functionals J_n defined on X with values in $\mathbb{R}$, one can extract a subsequence that Γ-converges.*

Let's see the link with the γ-convergence. We consider the Dirichlet problem for the Laplacian, let $\Omega \subseteq D$ with D an open bounded set in $\mathbb{R}^n$, for every function $f \in H^{-1}(\Omega)$ exists a solution in the sense of distributions for the system

$$\begin{cases} -\Delta u = f & \text{in } \Omega \\ u|_{\partial\Omega} = 0. \end{cases}$$

This solution can be characterized as the unique minimum of the functional J defined in $H_0^1(\Omega)$

$$J_\Omega(\nu) = \frac{1}{2}\int_\Omega |\nabla \nu|^2 - \langle f, \nu \rangle_{H^{-1}(\Omega), H_0^1(\Omega)}.$$

If we consider a sequence of domains Ω_n, one cannot directly consider the Γ-convergence of the sequence of functionals J_n since these are not defined on the same space, and if we see them as functional defined in $H_0^1(D)$ we lose information because we want that the solution belongs to $H_0^1(\Omega)$.

We introduce the indicator of the set $H_0^1(\Omega)$, defined by

$$\forall \nu \in H_0^1(D) \qquad \chi_{H_0^1(\Omega)}(\nu) = \begin{cases} 0 & if \nu \in H_0^1(\Omega) \\ +\infty & otherwise. \end{cases}$$

Now we consider the functional

$$J_\Omega(\nu) = \frac{1}{2}\int_D |\nabla \nu|^2 - \int_D f\nu \, dx + \chi_{H_0^1(\Omega)}(\nu).$$

Then we have the following proposition

Proposition 2.7. *A sequence of open sets Ω_n γ-converges to Ω if and only if, for any $f \in L^2(D)$, J_{Ω_n} Γ-converges to J_Ω.*

Proof. J_{Ω_n} Γ-converges to J_Ω implies the γ-convergence follows from the theorem by De Giorgi that states the convergence of the minima.

We divide the demonstration of the other implication in two parts. First we want to prove the point (i) of the definition of Γ-convergence. From Proposition 2.3. the γ-convergence is equivalent to the Mosco-convergence of $H_0^1(\Omega_n)$ to $H_0^1(\Omega)$. Let us take a sequence ν_n converging to ν strongly in $H_0^1(D)$. If $\nu_n \notin H_0^1(\Omega_n)$ except for finite number of n we have that $J_{\Omega_n}(\nu_n)$ is $+\infty$ so the result is obvious. Otherwise, $\nu_{n_k} \in H_0^1(\Omega_{n_k})$ converges to ν and for the second property of the Mosco convergence $\nu \in H_0^1(\Omega)$. So the third term of the functional is always zero. Then from the strong convergence of ν_{n_k} to ν we have $\lim_{k\to+\infty} J_{\Omega_{n_k}}(\nu_{n_k}) = J_\Omega(\nu)$.

To prove the point (ii) of the definition of the Γ-convergence we consider $\nu \in H_0^1(D)$. If $\nu \notin H_0^1(\Omega)$, $J_\Omega(\nu) = +\infty$ so it is obvious. If $\nu \in H_0^1(\Omega)$ for the first property of the Mosco convergence there exists a sequence $\nu_n \in H_0^1(\Omega_n)$ that strongly converges to $\nu \in H_0^1(D)$. We already know that $\nu \in H_0^1(\Omega)$ so we conclude as before that $\lim_{n\to+\infty} J_{\Omega_n}(\nu_n) = J_\Omega(\nu)$.

□

2.2 Relaxation for the Dirichlet problem

We consider a generic frame, what does it mean to relax a minimization problem?

Let us suppose that we want to solve an optimization problem like $\min_{x\in X} J(x)$, and that we are not able to prove existence of a solution (possibly because there is no solution). Then we try to embed X in a larger space X^* and we introduce a

new functional J^* defined on X^* that extends J in such a way that the following conditions are satisfied:

1) the relaxed problem $\min_{x^* \in X^*} J^*(x^*)$ has a solution;

2) $\inf_X J = \min_{X^*} J^*$;

3) the minimizing sequences of the initial problem converge (up to a subsequence) to a solution of the relaxed problem;

4) if the initial problem has a solution, then it is also a solution of the relaxed problem.

Usually, the relaxed space X^* is a completion of X in such a way that X is dense in X^*. In order to get the first condition above, it is natural to choose a space X^* that is compact and a functional J^* that is lower semicontinuous. Then it is possible to get conditions 2), 3) and 4) by imposing the following properties:

- if $(x_n) \subseteq X$ converge to $x^* \in X^*$, then $J^*(x^*) \leq \liminf J(x_n)$;
- for any $x^* \in X^*$, there exists a sequence x_n of X that converge to x^* and such that $J^*(x^*) = \liminf J(x_n)$.

A good choice for the relaxed functional J^* is the lower semicontinuous envelope of J defined by

$$J^*(x^*) := \inf\{\liminf J(x_n) \ \mid \ x_n \to x^*\}.$$

In this case, it is clear that the two conditions above are fulfilled.

Let's now consider our minimization problem, we have a solution of the Dirichlet problem $-\Delta u = f$ where $u \in H_0^1(\Omega)$ and a functional J that depends indirectly on Ω because is a function of u_Ω. So the functional can be seen as a two variables functional $J(\Omega, u_\Omega)$. Now the minimization problem is of the form

$$\min_{\Omega \in \mathcal{U}} J(\Omega, u_\Omega),$$

where Ω belongs to a certain set of the admissible shapes $\mathcal{U}$ and u_Ω is the solution of the associed Dirichlet equation.

As we have said above the solution u_Ω of the Dirichlet problem can be seen as the minimum of the functional

$$J_\Omega(\nu) = \frac{1}{2}\int_\Omega |\nabla \nu|^2 - \int_D f\nu \, dx + \chi_{H_0^1(\Omega)}(\nu).$$

To not be confused with the given initial functional J to be minimized, we recall this other functional G and we see it as a two variables functional

$$G(\Omega, \nu) = \frac{1}{2} \int_{\Omega} |\nabla \nu|^2 - \int_D f\nu \; dx + \chi_{H_0^1(\Omega)}(\nu),$$

and the solution u_Ω of the Dirichlet problem belongs to the $\text{argmin} G(\Omega, \cdot)$.

By now we minimize the problem searching a minimizer Ω in the set $\mathcal{U}$, if we see the functional as a two variables function we can search a minimizer that is a pair (Ω, u_Ω) in the set of the admissible pairs, that is of the form

$$\mathcal{A} = \{(\Omega, \nu) \;\; | \;\; \Omega \in \mathcal{U}, \nu \in \text{argmin} G(\Omega, \cdot)\}.$$

For what we have said above we know that a sequence of subset Ω_n γ-converges to Ω if and only if $G(\Omega_n, \cdot)$ Γ-converges to $G(\Omega, \cdot)$.

We now want to relax the problem, first of all we want to wander the set of the admissible shape to make it compact with respect to the γ-convergence, and then relax the functional J^* to have the properties described above.

We have that the shape Ω belongs to a set $\mathcal{U}$, we set the completion of $\mathcal{U}$ with respect to the γ-convergence and we call it $\hat{\mathcal{U}}$.

First of all we introduce the relaxed state functional $\hat{G}$, we have

$$\hat{G}(\hat{\Omega}, \cdot) = \Gamma \lim_{\Omega \to \hat{\Omega}} G(\Omega, \cdot).$$

We have that the state functionals $G(\Omega, \cdot)$ belong to the set $S_\Psi(H_0^1(D))$ defined as above, and we define a map

$$\Gamma_G : \mathcal{U} \to S_\Psi(H_0^1(D)) \qquad \text{such that} \qquad \Gamma_G(\Omega) = G(\Omega, \cdot),$$

that is one-to-one and an isometry.

We can extend this map as the unique isometry $\hat{\Gamma}_G$ that extends Γ_G, more precisely

$$\hat{\Gamma}_G(\hat{\Omega}) = \Gamma \lim_{n \to +\infty} \Gamma_G(\hat{\Omega}_n),$$

where $\hat{\Omega}_n$ is a sequence that γ-converges to $\hat{\Omega}$.

So we have that the relaxed state functional is

$$\hat{G}(\hat{\Omega}, \cdot) = \hat{\Gamma}_G(\hat{\Omega}).$$

For the proposition 1.42. we have that $\hat{\Omega}_n$ γ-converges to $\hat{\Omega}$ if and only if $\hat{G}(\hat{\Omega}_n, \cdot)$ Γ-converges to $\hat{G}(\hat{\Omega}, \cdot)$.

We have the following theorem:

Theorem 2.8. *$\hat{\mathcal{U}}$ is a compact metric space with respect to the γ-convergence.*

Proof. Since $\hat{\Gamma}_G$ is an isometry $\hat{\Gamma}_G(\hat{\mathcal{U}})$ is a complete subspace of the compact metric space $S_\Psi(H_0^1(D))$ that is compact. So $\hat{\Gamma}_G(\hat{\mathcal{U}})$ is compact and using again that $\hat{\Gamma}_G$ is an isometry also $\hat{\mathcal{U}}$ is compact. □

The set of the admissible pairs becomes of the form

$$\hat{\mathcal{A}} = \{(\hat{\Omega}, \nu) \ \mid \ \hat{\Omega} \in \hat{\mathcal{U}}, \nu \in \operatorname{argmin}\hat{G}(\hat{\Omega}, \cdot)\}.$$

We now define the relaxed state functional, for a pair $(\hat{\Omega}, u) \in \hat{\mathcal{A}}$ we set

$$\hat{J}(\hat{\Omega}, u) = \inf\{\liminf_{n\to+\infty} J(\Omega_n, u_n) \ \mid \ \Omega_n \to \hat{\Omega} \text{ in } \hat{\mathcal{U}},\ u_n \to u \text{ in } H_0^1(D),\ u_n \in \operatorname{argmin} G(\Omega_n, \cdot)\}. \tag{2.1}$$

Therefore the relaxed optimal control problem can be written in the form

$$\min\{\hat{J}(\hat{\Omega}, u) \ \mid \ \hat{\Omega} \in \hat{\mathcal{U}},\ y \in H_0^1(D),\ u \in \operatorname{argmin} G(u_n, \cdot)\}.$$

If the cost functional J depends only on the state variable $u \in H_0^1(D)$ we have that $J = \hat{J}$.

We have the following theorem

Theorem 2.9. *Under the assumptions above, the relaxed problem admits at least a solution $(\hat{\Omega}, u) \in \hat{\mathcal{A}}$. Moreover, the infimum of the original problem coincides with the minimum of the relaxed problem. Finally, if (Ω_n, u_n) is a minimizing sequence for the original problem, then there exists a subsequence converging to a solution $(\hat{\Omega}, u)$ of the relaxed problem.*

Proof. This is a consequence of the direct method of Calculus of Variations. Let us take a minimizing sequence Ω_n in $\mathcal{U}$, since it is compact there exists a set $\hat{\Omega}$ such that $\Omega_n \xrightarrow{\gamma} \hat{\Omega}$ in $\hat{\mathcal{U}}$. From Proposition 2.7. we know that $G(\Omega_n, \cdot)$ Γ-coverge to $G(\hat{\Omega}, \cdot)$ and for the theorem of De Giorgi also the sequence of the minima u_{Ω_n} converges to the minimum $u_{\hat{\Omega}}$. This proves the continuity of the solution for the Dirichlet problem. Since we have defined the relaxed functional as in (2.1), it is lower semicontinuous so $(\hat{\Omega}, u_{\hat{\Omega}})$ is the searched minimum. From the definition of the relaxed functional it follows that the conditions (2), (3), (4) imposed for the relaxed problem are satisfied. □

2.2.1 Necessary conditions of optimality

In this section we want to show the formulation of the relaxed form for a Dirichlet problem. We set the problem defining the set of the admisssible shapes and the functional $\mathcal{J}$. We consider as set of the admissible shapes

$$\mathcal{A}(D) = \{\Omega \subseteq D \ \mid \ \Omega \ \textit{quasi-open}\},$$

where D is a bounded open set in $\mathbb{R}^n$. As functional $\mathcal{J}$ we consider a functional of the form

$$\mathcal{J}(\Omega) = \int_D j(x, u_\Omega(x))dx, \tag{2.2}$$

with $j : \Omega \times \mathbb{R} \to \mathbb{R}$ such that

$$\begin{aligned} &i)\ g(x,\cdot) \text{ measurable in } x, \forall s, \\ &ii)\ g \text{ continuous in } s \text{ for a.e.} x \in \Omega, \\ &iii)\ \forall (x,s) \quad |j(x,s)| \leq a(x) + b|s|^q \text{ for } a \in L^1(D), b \in \mathbb{R} \text{ and } q \in [1, \frac{2n}{n-2}). \end{aligned} \tag{2.3}$$

To relax the problem we have first to enlarge the set of the possible shapes considering its completion with respect to the γ-convergence and then relax the functional $\mathcal{J}$.

We can also consider the set

$$\mathcal{A}(D) = \{\Omega \subseteq D \ \mid \ \Omega \ \textit{open}\},$$

that is contained in the first one and has the same completion for the γ-convergence that we will develop above.

Completion of $\mathcal{A}(D)$

We first define the set of measures $\mathcal{M}_0(D)$

Definition 2.10. *We define $\mathcal{M}_0(D)$ to be the set of all nonnegative Borel measures ν on D, possibly $+\infty$ valued, such that*

i) $\mu(B) = 0$ for every Borel set $B \subseteq D$ with $\operatorname{cap}(B) = 0$,

ii) $\mu(B) = \inf\{\mu(U) : U\}$ quasi-open, $B \subseteq U\}$ for every Borel set $B \subseteq D$.

For example, if $n - 2 < \alpha \leq n$ the α-dimensional Hausdorff measure $\mathcal{H}^\alpha$ belongs to $\mathcal{M}_0(D)$. We fix a bounded open set $D \subseteq \mathbb{R}^n$ ans we consider a subset $\omega \subseteq D$,

another example of measure of the class $\mathcal{M}_0(D)$ is, for every open set $\omega \subseteq D$, the measure μ_ω defined by

$$\mu_\omega(A) = \begin{cases} 0 & if\ \mathrm{cap}_D(A \setminus \omega) = 0 \\ +\infty & if\ \mathrm{cap}_D(A \setminus \omega) > 0. \end{cases} \tag{2.4}$$

In the same hypothesis as above for ω and D, we recall the Dirichlet problem for the Laplacian

$$\begin{cases} -\Delta u = f \quad in\ \omega \\ u \in H_0^1(\omega), \end{cases} \tag{2.5}$$

where $f \in H^{-1}(D)$. We call u_ω the solution of that system and we can write it by the variational formulation, $u_\omega \in H_0^1(\omega)$ is the unique solution of

$$\int_\omega \nabla u_\omega \nabla \varphi = \int_\omega f\varphi \qquad \forall \varphi \in H_0^1(\omega).$$

We now consider a partial differential equation of the form

$$\begin{cases} -\Delta u + u\mu = f\ in\ D \\ u \in H_0^1(D), \end{cases} \tag{2.6}$$

with $\mu \in \mathcal{M}_0(D)$, and we call u_μ its solution. For the (1.8) of Proposition 1.23. u_μ is the unique minimizer of the functional

$$F(\nu) = \frac{1}{2}\int_D |\nabla \nu|^2 + \frac{1}{2}\int_D \nu^2 d\mu - \int_D f\nu, \tag{2.7}$$

so it belongs to the subspace $H_0^1(D) \cap L^2(D,\mu)$, where $L^2(D,\mu)$ denotes the space of measurable functions that are square integrable for the measure μ. A clarification, with $\int_D \nu^2 d\mu$ in fact we consider $\int_D \tilde{\nu}^2 d\mu$ with $\tilde{\nu}$ the quasi-continuous representative of ν.

We can call this space $X_\mu(D) = H_0^1(D) \cap L^2(D,\mu)$ and we can endow the space it with the norm

$$\|u\|_{X_\mu(D)} = \left(\int_D |\nabla u|^2 dx + \int_D u^2 d\mu\right)^{\frac{1}{2}}$$

which comes from the scalar product

$$(u,v)_{X_\mu(D)} = \int_D \nabla u \nabla v dx + \int_D uv d\mu$$

With the scalar product above the space $X_\mu(D)$ becomes a Hilbert space.

Theorem 2.11. *$X_\mu(D)$ is a Hilbert space.*

Proof. It is enough to prove the completeness. Let u_n be a Cauchy sequence in $X_\mu(D)$. Then u_n is a Cauchy sequence both in $H_0^1(D)$ and in $L^2(D,\mu)$. Therefore u_n converges to a function u in $H_0^1(D)$ and to a function v in $L^2(D,\mu)$. By theorem 1.22. there exists a subsequence u_{n_k} that converges to u q.e. in D. Since $\mu \in \mathcal{M}_0(D)$ vanishes on all sets with capacity zero, u_{n_k} converges to u μ-a.e. in D. On the other hand v in $L^2(D,\mu)$ so a further subsequence of u_{n_h} converges to v μ-a.e. in D hence $v = u$ μ-a.e. in D and ,therefore, $u \in X_\mu(D)$. u_n converges up to a subsequence to u both in $H_0^1(D)$ and in $L^2(D,\mu)$, this implies that it converges to u in $X_\mu(D)$. □

We can also characterize the solution $u_\mu \in H_0^1(D)$ of problem (2.6) by a variational formulation (Proposition 1.23 (1.6)):

$$\int_D \nabla u_\mu \nabla \nu dx + \int_D u_\mu \nu d\mu = \int_D f\nu \qquad \forall \nu \in X_\mu(D). \tag{2.8}$$

We now consider, for every open set $\omega \subseteq D$ the measure μ_ω defined above in (2.4). Taking account the characterization of $H_0^1(\omega)$ given in Theorem 1.21. we have that $\nu \in H_0^1(\omega)$ if and only if its quasi-continuous representative $\tilde{\nu} = 0$ quasi-everywhere on $D \setminus \omega$. This implies that

$$\nu \in H_0^1(\omega) \Rightarrow \mathrm{cap}_D([\nu \neq 0] \setminus \omega) = 0 \Rightarrow \int_D \nu^2 d\mu_\omega = 0$$

and,

$$\nu \in H_0^1(D) \setminus H_0^1(\omega) \Rightarrow \mathrm{cap}_D([\nu \neq 0] \setminus \omega) > 0 \Rightarrow \int_D \nu^2 d\mu_\omega = +\infty.$$

The solution u_ω of the equation (2.5) coincides with the solution u_{μ_ω} of the equation (2.6) with the measure μ_ω defined in (2.4). If we consider the variational formulation of problem (2.5), u_ω is the unique minimizer of the functional

$$G(\nu) = \frac{1}{2}\int_D |\nabla \nu|^2 - \int_D f\nu,$$

and u_{μ_ω} is the unique minimizer of the functional F in (2.7). From the properties above we have that, if $\nu \in H_0^1(\omega)$ the two functional concides $F(\nu) = G(\nu)$, and if $\nu \notin H_0^1(\omega)$, $F(\nu) = +\infty$ so the minimizer is a function in $H_0^1(\omega)$ and coincides with the minimizer of F.

We identify any open set ω with the nonnegative Borel measure μ_ω, this last property allows us to embed $\mathcal{A}(D)$ in $\mathcal{M}_0(D)$ by the map $\omega \to \mu_\omega$. The equation (2.6) is the state equation of the relaxed optimal control problem associated with the initial state equation (2.5), in particular, a solution of the relaxed problem is a solution of the original problem if it is a measure $\mu \in \mathcal{M}_0(D)$ of the form $\mu = \mu_\omega$, and ω is the searched minimizer.

An important fact is that the class of the measures of the form μ_ω, with ω an open subset of D, is dense in $\mathcal{M}_0(D)$.

We define the γ-convergence in $\mathcal{M}_0(D)$,

Definition 2.12. *We say that a sequence of measures μ_n of $\mathcal{M}_0(D)$ γ-converges to a measure $\mu \in \mathcal{M}_0(D)$ if, for any $f \in H^{-1}(D)$, the sequence of solutions u_{μ_n} of problem (2.6) weakly converges in $H_0^1(D)$ to u_μ.*

We have the following theorem

Theorem 2.13. *(Dal Maso-Mosco)*

The set $\mathcal{M}_0(D)$ is a realization of the completion of $\mathcal{A}(D)$ for γ-convergence. Moreover $\mathcal{M}_0(D)$ is compact for γ-convergence.

Relaxation of the functional

From the paragraph above we have that the relaxed state equation is

$$\begin{cases} -\Delta u + u\mu = f \text{ in } D \\ u \in H_0^1(D), \end{cases}$$

with μ a measure in $\mathcal{M}_0(D)$. The solution u_μ has to be intended in the weak sense: $u_\mu \in X_\mu(D)$ and

$$\int_D \nabla u_\mu \nabla \nu dx + \int_D u_\mu \nu d\mu = \int_D f\nu \qquad \forall \nu \in X_\mu(D).$$

We have the following lemma

Lemma 2.14. *For any measurable set $\omega \subseteq D$, the functional defined on $H_0^1(D)$ by*

$$\nu \to \int_\omega j(x, \nu(x))dx,$$

is continuous for the weak convergence of $H_0^1(D)$ and for the strong convergence in $L^2(D)$.

Since the γ-convergence in the completion $\mathcal{M}_0(D)$ coincides with the weak convergence in $H_0^1(D)$ of the solution u_μ and since,for the lemma above, the functional $\mathcal{J}$ is continuous, the corresponding relaxed functional remains the same. The relaxed problem becomes

$$\min_{\mu\in\mathcal{M}_0(D)}\{\int_D (j(x,u_\mu)dx \;\mid\; u_\mu \in H_0^1(D) \text{ solution of } (2.6)\}. \tag{2.9}$$

In particular, the relaxed problem fulfills the hypothesis of Theorem 2.9. so it admits a minimizer.

We now want to find a way to characterize the minimum by deducing some necessary conditions for optimality.

We work with the space of functions $X_\mu(D)$ that can be embedded in $H_0^1(D)$ by the identity map, so the dual space $H^{-1}(D)$ can be considered as a subspace of the dual space $X'_\mu(D)$. For simplicity we can consider the resolvent operator $R_\mu : X'_\mu(D) \to X_\mu(D)$ which associates to every $f \in X'_\mu(D)$ the unique solution of the (2.6), so we have $u_\mu = R_\mu(f)$.

In this section we obtain two optimality conditions for a solution $\mu \in \mathcal{M}_0(D)$ of the relaxed optimization problem. The general method to prove these results consists in computing the limit

$$\lim_{\varepsilon\to 0}\frac{\mathcal{J}(u_{\mu_\varepsilon})-\mathcal{J}(u_\mu)}{\varepsilon},$$

for a suitable family μ_ε of perturbations of μ.

Let us fix a function $f \in X'_\mu(D)$ and we assume that the function $j : D \times \mathbb{R} \to \mathbb{R}$ satisfies the conditions expressed at the beginning of the paragraph and, in addition, the following conditions.

$i)$ $j(x,\cdot)$ is differentiable and its derivative $j_s(x,\cdot)$ is continuous on $\mathbb{R}$ for a.e. x
$ii)$ $j_s(x,\cdot) =$ is $\mathcal{L}^n$-measurableon D for every s
$iii)$ $|j_s(x,s)| \le a(x) + c|s|$ for $a \in L^2(D)$ and $c \in \mathbb{R}$.

The map $\mathcal{J}(u) = \int_D j(x,u(x))dx$, is differentiable on $X_\mu(D)$ and that its differential $\mathcal{J}'$ is given by

$$\langle J'(u), v\rangle = \int_D j_s(x,u)v dx \qquad \forall u, v \in X_\mu(D)$$

where $\langle\cdot,\cdot\rangle$ is the pairing between $X_\mu(D)$ and its dual.

Given $\mu \in \mathcal{M}_0(D)$, let u be the solution of the problem (2.6),

$$u \in X_\mu(D) \qquad -\Delta u + u\mu = f \qquad in\ X'_\mu(D). \tag{2.10}$$

and ν be the solution of the adjoint problem

$$\nu \in X_\mu(D) \qquad -\Delta \nu + \nu\mu = j_s(x,u) \qquad in\ X'_\mu(D). \tag{2.11}$$

We have the following theorem

Theorem 2.15. *Let $\mu \in \mathcal{M}_0(D)$ be a minimum point of the relaxed optimization problem* (2.9)*, and let u and v be the solutions of* (2.10) *and* (2.11)*.*

Then $uv \leq 0$ q.e in D.

Proof. Let us fix $\varphi \in L^\infty(D)$ with $\varphi \geq 0$ a.e. in D. For every $\varepsilon \geq 0$, let $\mu_\varepsilon \in \mathcal{M}_0(D)$ be the measure defined by $\mu_\varepsilon = \mu + \varepsilon\varphi\mathcal{L}^n$, and let u_ε be the corresponding solution of the problem

$$u_\varepsilon \in X_{\mu_\varepsilon}(D) \qquad -\Delta u_\varepsilon + u_\varepsilon\mu_\varepsilon = f \qquad in X'_{\mu_\varepsilon}(D). \tag{2.12}$$

Note that if $\varepsilon = 0$, $\mu_\varepsilon = \mu$ and $u_\varepsilon = u$. Since φ is bounded, we have $X_\mu(D) = X_{\mu_\varepsilon}(D)$ with equivalent norms. If we consider the variational formulation u_ε satisfies

$$\int_D \nabla u \nabla \nu dx + \int_D u\nu d(\mu + \varepsilon\varphi\mathcal{L}^n) = \int_D f\nu \qquad \forall \nu \in X_\mu(D),$$

then

$$\int_D \nabla u \nabla \nu dx + \int_D u\nu d\mu = \int_D f\nu dx - \int_D u\nu\varepsilon\varphi dx \qquad \forall \nu \in X_\mu(D).$$

Therefore, u_ε satisfies

$$u_\varepsilon \in X_{\mu_\varepsilon}(D) \qquad -\Delta u_\varepsilon + u_\varepsilon\mu = f + \varepsilon\varphi u \qquad in\ X'_{\mu_\varepsilon}(D),$$

or equivalently $u_\varepsilon = R_\mu(f - \varepsilon\varphi u_\varepsilon) = R_\mu(f) - R_\mu(\varepsilon\varphi u_\varepsilon)$.

We have that the map $\varepsilon \to u_\varepsilon$ is differentiable at $\varepsilon = 0$ and

$$\frac{du_\varepsilon}{d\varepsilon}|_{\varepsilon=0} = -R_\mu(\varphi u),$$

and so

$$\frac{d\mathcal{J}(u_\varepsilon)}{d\varepsilon}|_{\varepsilon=0} = -\int_D j_s(x,u)R_\mu(\varphi u)dx.$$

On the other hand, the optimality of μ gives that the derivative above has to be nonnegative, so that we obtain

$$\int_D j_s(x,u)R_\mu(\varphi u)dx \leq 0 \qquad \forall \varphi \in L^\infty(D), \varphi \geq 0.$$

By the symmetry of the resolvent operator R_μ we can also write

$$\int_D R_\mu(j_s(x,u))\varphi u dx \leq 0 \qquad \forall \varphi \in L^\infty(D), \varphi \geq 0,$$

which gives, since $v = R_\mu(j_s(x,u))$ and φ is arbitrary,

$$uv \leq 0 \text{ a.e. in } D.$$

Finally, since uv is finely continuous q.e. in D, and every nonempty finely open set has positive Lebesgue measure, we obtain that $uv \leq 0$ *q.e.* in D. □

Theorem 2.16. *Let $\mu \in \mathcal{M}_0(D)$ be a minimum point of the relaxed optimization problem* (2.9)*, and let u and v be the solutions of* (2.10) *and* (2.11)*.*

Then $uv = 0$ $\mu - a.e$ in D.

Proof. For every $0 \leq \varepsilon < 1$, let $\mu_\varepsilon \in \mathcal{M}_0(D)$ be the measure defined by $\mu_\varepsilon = (1-\varepsilon)\mu$, and let u_ε be the corresponding solution of the problem

$$u_\varepsilon \in X_{\mu_\varepsilon}(D) \qquad -\Delta u_\varepsilon + u_\varepsilon \mu_\varepsilon = f \qquad in X'_{\mu_\varepsilon}(D). \tag{2.13}$$

Using the same argument as above, u_ε satisfies

$$u_\varepsilon \in X_{\mu_\varepsilon}(D) \qquad -\Delta u_\varepsilon + u_\varepsilon \mu = f + \varepsilon \mu u \qquad in X'_{\mu_\varepsilon}(D),$$

or equivalently $u_\varepsilon = R_\mu(f + \varepsilon \mu u_\varepsilon)$.

We have that the map $\varepsilon \to u_\varepsilon$ is differentiable at $\varepsilon = 0$ and

$$\frac{du_\varepsilon}{d\varepsilon}|_{\varepsilon=0} = R_\mu(\mu u),$$

and so

$$\frac{d\mathcal{J}(u_\varepsilon)}{d\varepsilon}|_{\varepsilon=0} = \int_D j_s(x,u)R_\mu(\mu u)dx.$$

On the other hand, the optimality of μ gives that the derivative above has to be nonnegative, so that we obtain

$$\int_D j_s(x,u)R_\mu(\mu u)dx \leq 0.$$

By the symmetry of the resolvent operator R_μ we can also write

$$\int_D R_\mu(j_s(x,u))u\mu dx \leq 0,$$

which gives, since $v = R_\mu(j_s(x,u))$,

$$uv \geq 0 \ \mu - a.e. \text{ in } D.$$

For the Theorem 1.51. above $uv \leq 0$ $q.e.$ in D, so $uv \leq 0$ $\mu - a.e.$ in D and we obtain that

$$uv = 0 \ \mu - a.e. \text{ in } D.$$

□

Relaxation phenomenon

We resume the example of nonexistence of a minimizer given in Chapter 1., we show that the minimizer of the problem is a measure $\mu \in \mathcal{M}_0(D)$ and so is not a shape and thus not a solution for the original problem. This is called 'relaxation phenomenon'.

We recall the problem. We consider a bounded open set $\Omega \in \mathbb{R}^n$ and a subset $\omega \subseteq \Omega$. We have the functional

$$\mathcal{J}(\omega) = \int_\Omega |y_\omega - c|^2 dx, \tag{2.14}$$

with c is a fixed constant, the set of possible shapes

$$\mathcal{A} = \{\omega \subset \Omega \ \mid \ \omega \text{ open}\},$$

and $y_\omega \in H_0^1(\omega)$ the solution of

$$\begin{cases} -\Delta y = 1 \quad in \ \omega \\ y|_{\partial\omega} = 0, \end{cases} \tag{2.15}$$

with $f \equiv 1 \in H^{-1}(\Omega)$. The optimization problem is

$$\min_{\omega \in \mathcal{A}} \mathcal{J}(\omega).$$

We will show that the minimizer in the case Ω is a ball, is $\mu = \frac{1}{c}|_{\overline{B(0,r_1)}} + \frac{k}{c}\mathcal{H}^{n-1}_{\partial B(0,r_1)}$ for k a constant that we will specify below.

This is a specific case of a more general one. We have u_ω is the solution of the Dirichlet problem

$$\begin{cases} -\Delta y = f \quad in\ \omega \\ y|_{\partial\omega} = 0, \end{cases}$$

with $f \in H^{-1}(\Omega)$ and $f \geq 0$ in Ω. This $\geq$ is understood either in the sense of distributions, or measures, or $H^{-1}(\Omega)$. And the functional $\mathcal{J}$ is of the form (2.2)

$$\mathcal{J} = \int_\Omega j(x, u_\omega)dx,$$

with j satisfying conditions (2.3).

We have shown above that the relaxed form for the problem $\min_{\omega\in\mathcal{A}} \int_\Omega j(x, u_\omega)dx$ is

$$\min_{\mu\in\mathcal{M}_0(\Omega)} \int_\Omega j(x, u_\mu)dx, \tag{2.16}$$

where u_μ is the solution of

$$\begin{cases} -\Delta y + y\mu = f \quad in\ \Omega \\ y|_{\partial\Omega} = 0. \end{cases}$$

We know that the relaxed functional admits a minimizer by Theorem 2.9.

We won't show in this work of thesis the complete characterization of the minimizer in the generic case, we give only some specific results that we will use to compute the explicit solution in our case.

There are some important results in the case where $j(x, \cdot)$ is differentiable and where Ω is a ball and the data are nonnegative and radially symmetric.

We assume that $j(x, \cdot)$ is smooth for a.e. $x \in \Omega$. More precisely, we assume that

$$\begin{aligned} &i)\ j_s(x, \cdot)\text{is differentiable for a.e.} x \in \Omega, \\ &ii)\ j_s(x, s) \text{ is a Carathéodory function,} \\ &iii)\ |g_s(x, s)| \leq a(x) + b|s| \qquad \text{for a.e. } x \in \Omega, \forall s. \end{aligned} \tag{2.17}$$

We can define the adjoint state as

$$v \in H_0^1(\Omega), \qquad -\Delta v = j_s(x, u)\text{in}\Omega.$$

Then we have the result:

Theorem 2.17. *Assume j satisfies conditions* (2.3) *and j_s conditions* (2.17). *If u is a minimizer to* (2.16) *and v is the adjoint state, then (u,v) satisfies the complementarity problem*

$$\Delta u + f \geq 0, \qquad v \leq 0 \text{ q.e. in } \Omega, \qquad \langle \Delta u + f, v \rangle = 0. \tag{2.18}$$

For $x \in \Omega$ we introduce the function Ψ defined by

$$\Psi(x) = sup\{s \ \mid \ j(x,s) \text{ is decreasing on } (-\infty, s)\}. \tag{2.19}$$

For $f \geq 0$ we have the following theorem

Theorem 2.18. *Assume that*

$$f \geq 0, \qquad f \neq 0 \text{ in any connected component of } \Omega \tag{2.20}$$
$$\inf_{\Omega'} \Psi > c > 0 \qquad \forall \Omega' \subset\subset \Omega.$$

Let u be a minimizer of (2.16). *Then*

$$\inf_{\Omega'} u > d > 0 \qquad \forall \Omega' \subset\subset \Omega$$

Therefore, the measure μ which satisfies that $u = u_\mu$ is uniquely determined by u and is a Radon measure.

We assume that

$$j_s(x,s) < 0 \ \forall s < \Psi(x), \quad \text{for a.e. } x \in \Omega, \tag{2.21}$$

$$j_s(x,s) > 0 \ \forall s > \Psi(x), \quad \text{for a.e. } x \in \Omega, \tag{2.22}$$

Now we assume that $\Omega = B(0,R)$, for a fixed radius R, and that f, Ψ and j are radially symmetric. More precisely, denoted by $\mathcal{O}(n)$ the group of orthogonal transformations ρ of $\mathbb{R}^n$, for any distribution S in $\mathcal{D}'(\Omega)$, ρS is the distribution defined by

$$\langle \rho S, \varphi \rangle = \langle S, \varphi \circ \rho \rangle \qquad \forall \varphi \in \mathcal{D}(\Omega).$$

Then we assume that

$$\rho f = f \qquad \forall \rho \in \mathcal{O}(n). \tag{2.23}$$

Moreover, for every s, we suppose that there exists a function $\tilde{j}(.,s)$ on $(0,R)$ such that

$$j(x,s) = \tilde{j}(|x|,s) \quad \forall s \in \mathbb{R}, \text{for a.e. } x \in \Omega. \tag{2.24}$$

As a consequence, the function Ψ defined in (2.19) satisfies

$$\Psi(x) = \tilde{\Psi}(|x|) \text{ for a.e. } x \in \Omega, \tag{2.25}$$

for some function $\tilde{\Psi}$ defined on (O, R).

We assume also that

$$\Psi \in H_0^1(\Omega), \qquad \Delta\Psi + f \geq 0 \; in\Omega. \tag{2.26}$$

and

$$\Psi \geq \eta > 0 \text{a.e. } in\Omega, \tag{2.27}$$

for η a positive constant.

We denote by u_0 the solution of

$$\begin{cases} -\Delta y = f \quad in\ \Omega \\ y|_{\partial\Omega} = 0, \end{cases} \tag{2.28}$$

The function u_0 is the solution of the relaxed state equation corresponding to $\mu = 0$.

Under all this assumption we are now able to write the following theorem

Theorem 2.19. *Assume $j(x, s)$ satisfies conditions* (2.3), (2.17), (2.21), (2.22), (2.24). *Assume Ψ defined by* (2.19) *satisfies* (2.26),(2.27) *and assume f satisfies* (2.23). *Then, if u is a radially symmetric minimizer of* (2.16)*:*

There exist r_1, r_2, with $0 \leq r_1 \leq r_2 < R$, such that

$$\tilde{u}(r) = \tilde{\Psi}(r) for\ 0 < r \leq r_1, \tag{2.29}$$

$$\tilde{u}(r) > \tilde{\Psi}(r) for\ r_1 < r \leq r_2, \tag{2.30}$$

$$\tilde{u}(r) < \tilde{\Psi}(r) for\ r_2 < r < R. \tag{2.31}$$

If $u \neq u_0$ solution of (2.28), *then*

$$0 < r_1 < r_2 < R.$$

In this case the measure $\mu \in \mathcal{M}_0(\Omega)$ such that $u = u_\mu$ is uniquely determined by

$$\mu = \frac{\lambda|_{B(0,r_1)}}{\Psi} + \frac{f|_{\partial B(0,r_1)}}{\tilde{\Psi}(r_1)} + \frac{\tilde{u}'(r_1^+) - \tilde{\Psi}(r_1^-)}{\Psi(\tilde{r}_1)} \mathcal{H}^{n-1}_{\partial B(0,r_1)} \tag{2.32}$$

where $\lambda = -\Delta\Psi + f$ is a Radon measure, $\mathcal{H}^{n-1}_{\partial B(0,r_1)}$ denotes the $(n-1)$-dimensional Hausdorff measure of $\partial B(0, r_1)$.

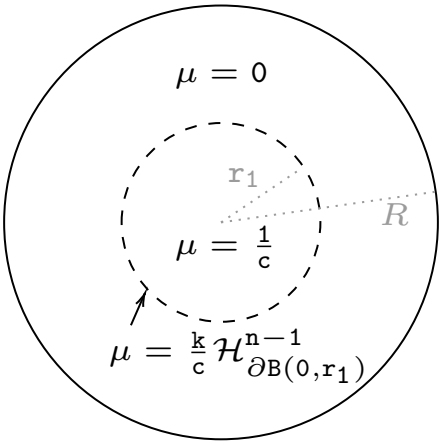


Figure 2.1

Now we can go back to our problem. In our case the function

$$j(x,s) = c, \qquad c = \text{constant},$$

so is radial and satisfies conditions (2.3), (2.17), (2.21), (2.22), (2.24). The function $f \equiv 1$ is constant and radial.

We rewrite here the functional $\mathcal{J}$

$$\mathcal{J}(\omega) = \int_\Omega |y_\omega - c|^2 dx.$$

We have to distingue some cases:

1. If $c \leq 0$ the solution is $y = 0$ in Ω,
2. if $c \geq \sup_\Omega u_0$ then the solution is $u = u_0$,
3. if $c < \sup_\Omega u_0$, u_0 is not a minimizer because the (u_0, v_0) doesn't satisfy (2.18), thus we can apply Theorem 2.19.. We conclude that for $0 < r_1 < r_2 < R$, there exist a unique minimizer such that

$$u = c \ in \ B(0, r_1),$$

and

$$\mu = \frac{1}{c}|_{\overline{B(0,r_1)}} + \frac{\tilde{u}'(r_1^+)}{c}\mathcal{H}^{n-1}_{\partial B(0,r_1)}. \tag{2.33}$$

2.3 Differentiating with respect to domains

We have functionals depending on varying domain and we want to derivate the functional with respect to the domain, trying to write optimality conditions of first and second order. The classical notion of differentiability requires a normed vector space framework, but there is no canonical one when dealing with a family of open subsets or domains of $\mathbb{R}^n$.

Let $E : \mathcal{O} \to X$, where $\mathcal{O}$ is a family of subsets of $\mathbb{R}^n$ and X is a normed vector space. A rather convenient approach is to consider the application $\theta \to \mathcal{E}(\theta) = E((I+\theta)(\Omega))$, where θ varies around 0 in a normed vector space Θ of applications from $\mathbb{R}^n$ into itself, or from a fixed holdall D to $\mathbb{R}^n$. We then speak of Fréchet differentiability in the classical sense for the application $\theta \in \Theta \to \mathcal{E}(\theta) \in X$. However, compute Fréchet derivatives can become rather technical and complex, so it is often more pleasant to compute derivatives with respect to a real variable. Thus, we should consider and analyze derivatives of functions $t \in [0, T) \to E(\Omega_t)$, where $\Omega_t = \Phi(t)(\Omega)$ and $\Phi_t \in \Theta$. A classical example consists in taking $\Phi(t) = I + t\theta$, where θ is a regular vector field from $\mathbb{R}^n$ into $\mathbb{R}^n$.

In general we work with functionals depending on the solution of a partial differential equation that depends on a varying domain, i.e. we should differentiate $t \to u_t \in H_0^1(\Omega_t)$. Avoiding the differentiation of the state equation with respect to the domain is particularly advantageous in shape problems. Two main techniques are used to get around this difficulty. The first is 'function space parametrization' that consists in transporting the functions on variable domains back onto the initial domain, where they can be compared. The second is 'function space embedding' that consists in constructing extensions of the domain dependent boundary value problems to a larger fixed domain, for instance $\mathbb{R}^n$.

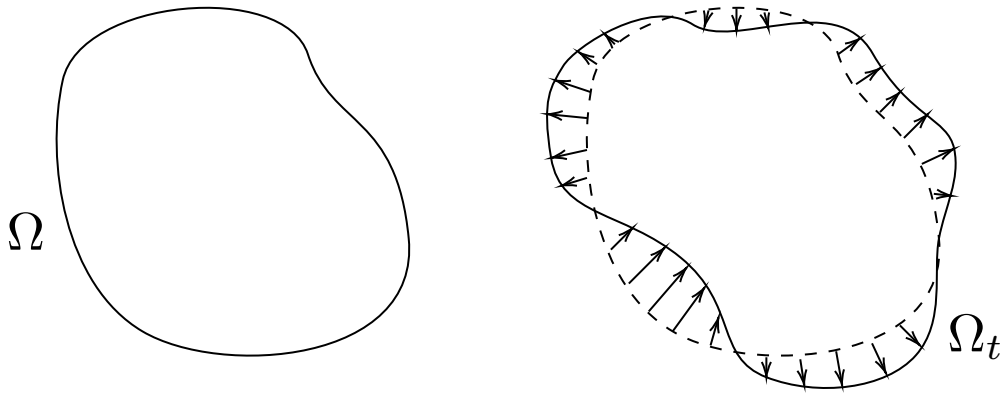


Figure 2.2: Perturbation given by a smooth vector field.

2.3.1 Shape derivative

We consider an open set $D \subseteq \mathbb{R}^n$ and a class of domain $\Omega \subseteq D$ that varies remaining always inside the holdall D, where D could be $\mathbb{R}^n$.

We consider the action of a class of diffeomorphisms associated to a velocity vector field V. We consider a velocity field

$$V : [0, T) \times \overline{D} \to \mathbb{R}^n, \tag{2.34}$$

that satisfies conditions

$$\forall x \in \overline{D}, \qquad V(\cdot, x) \in C([0, T); \mathbb{R}^n), \tag{2.35}$$

$$\exists c > 0,\ \forall x, y \in \overline{D}, \quad \|V(\cdot, y) - V(\cdot, x)\|_{C([0,T];\mathbb{R}^n)} \leq c|x - y|. \tag{2.36}$$

$$\forall t \in [0, T),\ \forall x \in \overline{D}, \quad V(t, x) \in L_D(x), \tag{2.37}$$

where $L_D(x)$ is the linear tangent space to D at the point $x \in \partial D$. We introduce the linear subspace

$$\mathrm{Lip}_L(D, \mathbb{R}^n) = \{\theta \in \mathrm{Lip}(D, \mathbb{R}^n)\ :\ \forall x \in \partial D,\ \theta(x) \in L_D(x)\} \tag{2.38}$$

Under the action of the velocity field V a set $\Omega \subseteq D$ is transformed into a new domain $\Omega_t(V) = \Phi_t(V)(\Omega)$ also contained in D. The transformation $\Phi_t : \overline{D} \to \overline{D}$ is defined as the flow associated to the vector field V, i.e.

$$\Phi_t(x) = \Phi(t, x), \qquad t \geq 0, x \in \overline{D},$$

where $\Phi(t, x)$ is the solution of

$$\frac{d}{dt}\Phi(t, x) = V(t, \Phi(t, x)), \qquad t \geq 0. \tag{2.39}$$

Definition 2.20. *Considering a velocity field V satisfying the conditions above we define the shape derivative at Ω in the direction V of the functional J as the limit, if exists,*

$$dJ(\Omega, V) = \lim_{t \searrow 0} \frac{J(\Omega_t(V)) - J(\Omega)}{t}. \tag{2.40}$$

For simplicity when $V(t, x) = \theta(x)$, $\theta \in \mathrm{Lip}_L(D, \mathbb{R}^n)$ is a velocity field we write $dJ(\Omega; \theta)$.

We say that a shape is critical if the first derivative vanishes, that is

$$\forall V \qquad dJ(\Omega, V) = 0.$$

And we have that a critical shape is minimal if the second derivative is positive.

First shape derivative for Domain Integrals

We consider functionals that are volume integrals over a bounded open domain Ω in $\mathbb{R}^n$. We consider a velocity field $V : [0,T) \times \mathbb{R}^n$ that satisfies conditions (2.35), (2.36), (2.37) and the associated family of transformations $\{\Phi_t : \ 0 \leq t < T\}$ where $\Phi_t(V)(\Omega) = \Omega_t(V)$. Further assume that $V \in C^0([0,T); C^1_{loc}(\mathbb{R}^n, \mathbb{R}^n)$ and that the Jacobian of $\Phi_t = \Phi(t, \cdot)$ is positive, i.e.

$$J_t = \det D\Phi_t = \det D_x \Phi(t, \cdot) > 0, \qquad (D\Phi_t)_{ij} = \partial j \Phi_i.$$

We note that Ψ_0 is the identity map, thus $J_0 = J_t(0,x) = I$.

Given a function f in $W^{1,1}_{loc}(\mathbb{R}^n)$, consider for $0 \leq t < T$ the volume integral

$$J(\Omega_t(V)) = \int_{\Omega_t(V)} f \ dx. \tag{2.41}$$

By the change of variables formula

$$J(\Omega_t(V)) = \int_{\Omega} f \circ \Phi_t J_t \ dx. \tag{2.42}$$

We have the following theorem

Theorem 2.21. *Let f be a function in $W^{1,1}_{loc}(\mathbb{R}^n)$. Assume that the vector field $V(t)$ satisfies conditions* (2.35), (2.36), (2.37).

i) For each $t \in [0,T)$ the map $f \to f \circ \Phi_t$ is from $W^{1,1}_{loc}(\mathbb{R}^n)$ to $W^{1,1}_{loc}(\mathbb{R}^n)$, and $\nabla(f \circ \Phi_t) = \ ^*(D\Phi_t)\nabla f \circ \Phi_t$.

ii) If $V \in C^0([0,T); C^1_{loc}(\mathbb{R}^n, \mathbb{R}^n)$ the map $t \to f \circ \Phi_t$ is well defined and for each t

$$\frac{\partial}{\partial t} f \circ \Phi_t = (\nabla f \cdot V(t)) \circ \Phi_t \in L^1_{loc}(\mathbb{R}^n).$$

iii) If $V \in C^0([0,T); C^1_{loc}(\mathbb{R}^n, \mathbb{R}^n)$ the map $t \to J_t$ is differentiable and for each t

$$\frac{\partial}{\partial t} J_t = [\mathrm{div} V(t)] \circ \Phi_t J_t \in C^0_{loc}(\mathbb{R}^n).$$

Then from Theorem 2.21. we can compute the shape derivative

$$\begin{aligned} dJ(\Omega; V) &= \frac{d}{dt} J(\Omega_t(V))|_{t=0} = \frac{d}{dt} \int_\Omega f \circ \Phi_t J_t \ dx \\ &= \int_\Omega \nabla f(\Psi(0,x)) \frac{\partial}{\partial t}(\Psi(t,x))|_{t=0} J_t(0,x) + f(\Psi(0,x)) \frac{\partial}{\partial t} J_t|_{t=0} dx \\ &= \int_\Omega \nabla f(x) V(0) + f(x) \mathrm{div} V(0) dx = \int_\Omega \mathrm{div} f V(0) dx. \end{aligned}$$

If Ω has a Lipschitzian boundary, then by Stokes's theorem

$$dJ(\Omega; V) = \int_{\partial\Omega} fV(0) \cdot n d\sigma. \tag{2.43}$$

We have the following theorem for function that depends on time

Theorem 2.22. *Assume that the velocity field $V(t)$ satisfies conditions* (2.35), (2.36), (2.37). *and $V \in C^0([0,T); C^1_{loc}(\mathbb{R}^n, \mathbb{R}^n)$. Given a function $f \in C([0,T); W^{1,1}_{loc}(\mathbb{R}^n)) \cap C^1([0,T); L^1_{loc}(\mathbb{R}^n))$. The first derivative of the functional*

$$J(\Omega_t(V)) = \int_\Omega f(t) \circ \Phi_t J_t \; dx,$$

at $t = 0$ is given by

$$dJ(\Omega_t(V)) = \int_\Omega f'(0) + \operatorname{div}(f(0)V(0)) dx. \tag{2.44}$$

If Ω has a Lipschitzian boundary,

$$dJ(\Omega; V) = \int_\Omega f'(0) + \int_{\partial\Omega} f(0)V(0) \cdot n d\sigma. \tag{2.45}$$

Hence if we consider a function $f(t,x)$ is indeed

$$\int_\Omega \frac{\partial}{\partial t} f(0,x) + \int_{\partial\Omega} (fV)(0,x) \cdot n d\sigma. \tag{2.46}$$

We give an example of using shape derivative, we consider the solution $y = y(\Omega)$ of the Dirichlet problem

$$-\Delta y = f, \qquad y = 0 \text{ in } \partial\Omega = \Gamma,$$

with $f \in H^{-1}(\Omega)$ a fixed function. The solution of this equation is the minimizing element in $H^1_0(\Omega)$ of the energy functional

$$E(\Omega, \varphi) = \int_\Omega \frac{1}{2} |\nabla\varphi|^2 - f\varphi \; dx.$$

We call J the shape functional equal to the minimum

$$J(\Omega) = \inf_{\varphi \in H^1_0(\Omega)} E(\Omega, \varphi) = \int_\Omega \frac{1}{2} |\nabla y|^2.$$

The shape derivative of the function is

$$dJ(\Omega; V) = -\frac{1}{2}\int_\Gamma |\frac{\partial y}{\partial n}|^2 V \cdot n \; d\Gamma.$$

We have that the minimizer must satisfy that $\forall V \; dJ(\Omega; V) = 0$, it yields to the extra boundary condition

$$\frac{\partial y}{\partial n} = 0 \qquad in \; \Gamma.$$

First shape derivative for Boundary Integrals

Now we consider, given $\varphi \in H^2_{loc}(\mathbb{R}^n)$ and a bounded open set Ω in $\mathbb{R}^n$ with Lipschitz boundary, the functional

$$J(\Omega) = \int_\Gamma \varphi d\Gamma,$$

Where $\Gamma = \partial\Omega$.

Given a velocity field $V(t)$ and the associated class of diffeomorphism Ψ_t as above, we consider

$$J(\Omega_t(V)) = \int_{\Gamma_t(V)} \varphi d\Gamma_t. \tag{2.47}$$

We want to bring back the integral from Γ_t to Γ. First we define the tangential Jacobian of Ψ_t on Γ

Definition 2.23. *Given T a C^1-diffeomorphism from $\mathbb{R}^n$ into itself, with $\Omega \to \Omega_T$ and $\Gamma \to \Gamma_T$. We call the tangential Jacobian of T on Γ the quantity*

$$J_\Gamma(T) = J_T|^*(DT)^{-1}n|, \tag{2.48}$$

where J_T is the Jacobian of T, $J_T = detDT$, n is the exterior unit normal vector to Γ and $^*(DT)^{-1}$ is the inverse of the cofactor matrix of DT where DT is the Jacobian matrix of T, $(DT)_{ij} = \frac{\partial T_i}{\partial x_j}$.

For the diffeomorphism Φ_t we note $J_\Gamma(\Phi_t) = w_t$, and we have that

$$J(\Omega_t(V)) = \int_{\Gamma_t(V)} \varphi d\Gamma_t = \int_\Gamma \varphi \circ \Phi_t w_t d\Gamma. \tag{2.49}$$

We have that the map $t \to w_t$ s differentiable and

$$w_t' = \text{div}V(0) - DV(0)n \cdot n,$$

hence the shape derivative is given by

$$dJ(\Omega;V) = \int_\Gamma \nabla\varphi V(0) + \varphi(\mathrm{div}V(0) - DV(0)n\cdot n)d\Gamma. \tag{2.50}$$

We introduce the tangential gradient and the tangential divergence

Definition 2.24. *Let $g : \Gamma \to \mathbb{R}$ be a function of class C^1 on Γ. We define its tangential gradient by*

$$\nabla_\Gamma g = \nabla G|_\Gamma - \frac{\partial G}{\partial n}n.$$

where $G \in C^1(\mathbb{R}^n, \mathbb{R})$ is an extension of g.

Definition 2.25. *Let $g : \Gamma \to \mathbb{R}^n$ be a function of class C^1 on Γ. We define its tangential divergence by*

$$\mathrm{div}_\Gamma g = \mathrm{div}G|_\Gamma - DG\ n\cdot n.$$

where $G \in C^1(\mathbb{R}^n, \mathbb{R}^n)$ is an extension of g.

we can rewrite the formula above condition defining the mean curvature of Γ by $H = div_\Gamma n$, and we obtain that

$$dJ(\Omega;V) = \int_\Gamma (\frac{\partial\varphi}{\partial n} + \varphi H)V(0)\cdot n\ d\Gamma. \tag{2.51}$$

Second-order shape derivative

In this thesis we don't study the second order shape derivative, but a general formulation is the following.

Given $f \in C^2_{loc}(\mathbb{R}^n)$ and a domain Ω of class C^2, consider the functional

$$J(\Omega_{t,s}(V,W)) = \int_{\Omega_{t,s}(V,W)} f\ dx,$$

where $\Omega_{t,s}(V,W) = \Phi_s(W)(\Omega_t(V)) = \Phi_s(W)(\Phi_t(V)(\Omega))$.

The objective is to compute

$$d^2J(\Omega;V,;W) = \frac{\partial}{\partial s}\left\{\frac{\partial}{\partial t}J(\Omega_{t,s}(V,W))|_{t=0}\right\}|_{s=0}. \tag{2.52}$$

In the case where $f = f(t,s)$ is only a function of time, with $f \in C^2([0,T) \times [0,T); H^2_{loc}(\mathbb{R}^n))$, $V \in C^0([0,T); C^2_{loc}(\mathbb{R}^n, \mathbb{R}^n)$ and $W \in C^0([0,T); C^1_{loc}(\mathbb{R}^n, \mathbb{R}^n)$, we have that

$$\begin{aligned}\frac{\partial}{\partial t} J(\Omega_{t,s}(V,W))|_{t=0} &= \int_{\Omega_s(W)} \frac{\partial f}{\partial t}(0,s) + \operatorname{div}(f(0,s)V(0))dx \\ &= \int_{\Omega_s(W)} \frac{\partial f}{\partial t}(0,s) + \int_{\Gamma_s(W)} f(0,s)V(0) \cdot n_s d\Gamma_s,\end{aligned}$$

and the second order derivative in $(t,s) = (0,0)$ is

$$\begin{aligned}d^2 J(\Omega; V, ; W) = \int_\Omega \frac{\partial}{\partial s}\left(\frac{\partial f}{\partial t}\right) dx + \operatorname{div}\left(\frac{\partial}{\partial s}V(0) + \frac{\partial}{\partial t}W(0)\right) \\ + \operatorname{div}[\operatorname{div}(fV(0)W(0)]dx = \int_\Omega \frac{\partial}{\partial s}\left(\frac{\partial f}{\partial t}\right) dx \\ + \int_\Gamma \left(\frac{\partial}{\partial s}V(0) + \frac{\partial}{\partial t}W(0)\right) \cdot n + \operatorname{div}(fV(0)W(0) \cdot n d\Gamma. \qquad (2.53)\end{aligned}$$

2.3.2 Topological derivative

The topological derivative measures the sensitivity of a given shape functional with respect to an infinitesimal singular domain perturbation, such as the insertion of holes. Let us consider a bounded domain $\Omega \subset \mathbb{R}^n$, with $n \geq 2$, which is subject to a nonsmooth perturbation confined in a small region $w_\varepsilon(\hat{x}) = \hat{x} + \varepsilon w$ of size ε, with $\overline{w_\varepsilon} \subset\subset \Omega$. In general the region $w_\varepsilon(\hat{x})$ is a small ball of radius ρ centered in $\hat{x}$, $B_\rho(\hat{x})$. Here, $\hat{x}$ is an arbitrary point of Ω and w is a fixed domain of $\mathbb{R}^n$. We consider domains Ω and $\Omega_\varepsilon = \Omega \setminus w_\varepsilon(\hat{x})$, and we consider a functional J that admits the following topological asymptotic expansion

$$J(\Omega_\varepsilon) = J(\Omega) + f(\varepsilon)g(\hat{x}) + o(f\varepsilon), \qquad (2.54)$$

where $f(\varepsilon)$ is a positive first order correction function of J. The function $\hat{x} \to g(\hat{x})$ is called the topological derivative of J at $\hat{x}$. Therefore, this derivative can be seen as a first order correction of $J(\Omega)$ to approximate $J\Omega_\varepsilon)$. In fact

$$\frac{J(\Omega_\varepsilon) - J(\Omega)}{f(\varepsilon)} = g(\hat{x}) + \frac{o(f(\varepsilon))}{f(\varepsilon)}. \qquad (2.55)$$

Passing to the limit for $\varepsilon \to 0$, we can define the topological derivative as

$$g(\hat{x}) = \lim_{\varepsilon \to 0} \frac{J(\Omega_\varepsilon) - J(\Omega)}{f(\varepsilon)}. \qquad (2.56)$$

If we consider a hole that is a ball, $B_\rho(\hat{x})$, then we have the same formulation but with a function of $f(\rho)$, and we have that $f(\rho) \sim \varepsilon^n$.

Let us consider a simple example to compute the topological derivative. We have a shape functional of the form

$$J(\Omega_\varepsilon) = \int_{\Omega_\varepsilon} g(x)dx,$$

where the function $g : \mathbb{R}^n \to \mathbb{R}$ is assumed to be continuous and $\Omega_\varepsilon = \Omega \setminus w_\varepsilon(\hat{x})$. Since $|w_\varepsilon| \to 0$ as $\varepsilon \to 0$, we have

$$J(\Omega) = \int_{\Omega} g(x)dx.$$

We are looking for an asymptotic expansion of the form (2.54), namely

$$\begin{aligned} J(\Omega_\varepsilon) &= \int_{\Omega_\varepsilon} g(x)dx + \int_{w_\varepsilon} g(x)dx - \int_{w_\varepsilon} g(x)dx = \int_{\Omega} g(x)dx - \int_{w_\varepsilon} g(x)dx \\ &= J(\Omega) - |w_\varepsilon| g(\hat{x}) + o(|w_\varepsilon|) \end{aligned} \tag{2.57}$$

For this example, the term $-g(\hat{x})$ is called the topological derivative of the shape functional J.

Chapter 3

Optimal control problem

In this chapters, we want to study an optimization problem where the functional $\mathcal{J}$ depends on the solution y_ω of an elliptic equation with source term that varies with the shape. In particular we have the following elliptic equation

$$\begin{cases} -\Delta y = \chi_\omega \\ y|_{\partial\Omega} = 0, \end{cases}$$

where the shape ω appears in the function and not in the variable boundary condition as in the previous problem. We use the theory of Optimal Control to solve the problem. In many cases we cannot find a minimizer but we can deduce some optimality conditions to characterize the minimum. In the last chapter we will study a specific problem of this form where we will apply the technique of relaxation and the result obtained from the Optimal Control theory.

3.1 Examples of existence of a minimum

First of all we give two examples of existence of the minimum for this latter kind of problem. In this two cases we can proceed using the Direct method of Calculus of Variation taking advantage of some compactness properties of the functional or of the set of possible shape.

We recall in this first paragraph the set of functions with bounded variation.

We consider an open bounded set $D \in \mathbb{R}^n$. We can define the variation of a given function $f \in L^1(D)$

Definition 3.1. *If $f \in L^1(D)$ we define*

$$Var(f) = \sup\{\int_D f \cdot div(\varphi)dx; \quad \varphi \in \mathcal{D}(D, \mathbb{R}^N) \quad \text{s.t.} \quad \|\varphi\|_\infty \leq 1\}.$$

If f is a Radon measure this is called the total variation and is usually noted by $Var(f) = \|f\|_1$ ((1.4)).

Now we can define the $BV - functions$.

Definition 3.2. *Let $f \in L^1(D)$ be a function. f is a function with bounded variation in D if the distributional derivative of f is a Radon finite measure, so*

$$\int_D f \nabla \varphi dx = -\int_D \varphi d\nabla f \qquad \forall \varphi \in C_c^\infty(D).$$

We denote the space of functions with bounded variation by $BV(D)$.

We have that f is a bounded variation function, $f \in BV(D)$, if and only if $Var(f) < \infty$ and $Var(f) = \|f\|_1 = |\nabla f|(D)$, where ∇f is the distributional derivative of f that is also a measure.

We define the $BV - norm$ by

Definition 3.3. *We define a norm in the space $BV(D)$ by*

$$\|f\|_{BV} = \|f\|_{L^1(D)} + \|f\|_1 . \tag{3.1}$$

Let Ω be an open set included in D, he $BV - norm$ of the characteristic function set on Ω is

$$\|\chi_\Omega\|_{BV} = \|\chi_\Omega\|_{L^1(D)} + \|\nabla \chi_\Omega\|_1 = |\Omega| + P_D(\Omega). \tag{3.2}$$

We can define also the weak-* convergence in $BV(D)$,

Definition 3.4. *Let $(u_n)_n \in BV(D)$ be a sequence. We say that a sequence $(u_n)_n \in BV(D)$ weak-* converges to a $u \in BV(D)$ if $u_n \to u$ in $L^1(D)$ and $\nabla u_n \overset{*}{\rightharpoonup} \nabla u$ for the weak-* topology of the space of the Radon finite measures on D.*

We have an important result for the compactness property of the $BV(D)$ space

Theorem 3.5. *Let $(u_n)_n \in BV(D)$ a bounded sequence, bounded with respect to the $BV - norm$, then there exist a subsequence that converges in the weak-* sense to $u \in BV(D)$.*

We consider an open bounded set $\Omega \subseteq \mathbb{R}^d$, and a measurable subset $\omega \subseteq \Omega$ and the set of the possible shapes

$$\mathcal{U}_L = \{\omega \subset \Omega \ \mid \ |\omega| \leq L|\Omega|\},$$

where L is a fixed real constant such that $0 < L < 1$.

We consider a functional $\mathcal{J}$ given by

$$\mathcal{J}(\omega) = \frac{1}{2} \|y_\omega - y_d\|^2_{L^2(\Omega)} + \alpha \|\chi_\omega\|_{BV},$$

for some fixed $\alpha > 0$ and $y_d \in L^2(\Omega)$. Using (3.2) we have

$$\mathcal{J}(\omega) = \frac{1}{2} \int_\Omega |y_\omega(x) - y_d(x)|^2 dx + \alpha(|\omega| + P_\Omega(\omega)).$$

For this problem a minimizer exists. The functional $\mathcal{J}$ is bounded from below by zero and the family of admissible ω such that $\mathcal{J}(\omega) < \infty$ is not empty. We know that the solution y_ω of the Dirichlet problem

$$\begin{cases} -\Delta y = \chi_\omega \\ y|_{\partial\Omega} = 0, \end{cases} \tag{3.3}$$

belongs to $H^1_0(\Omega)$ that is embedded in $L^2(\Omega)$ and the embedding is compact. So $\|y_\omega - y_d\|^2_{L^2(\Omega)}$ is finite for every ω in the admissible shapes and there exists a set ω that has finite volume and finite perimeter.

Let ω_n be a minimizing sequence and M be an upper bound of the sequence $\mathcal{J}(\omega_n)$. Then we have

$$\alpha(P_\Omega(\omega_n) + |\omega_n|) = \mathcal{J}(\omega_n) - \frac{1}{2} \|y_{\omega_n} - y_d\|^2_{L^2(\Omega)} < M.$$

So

$$P_\Omega(\omega_n) + |\omega_n| < \frac{M}{\alpha} \qquad \forall n$$

is a bounded sequence and this is exactly the $BV - norm$ of the characteristic function χ_{ω_n}. For the weak-* compactness of $BV(\Omega)$ we know the existence of ω and of a subsequence ω_{n_k} such that $\chi_{\omega_{n_k}} \to \chi_\omega$ in $L^1(\Omega)$. In particular the set $\omega \in \mathcal{U}_L$ because

$$|\omega_n| = \|\chi_{\omega_n}\|_{L^1(\Omega)} \to \|\chi_\omega\|_{L^1(\Omega)} = |\omega|,$$

and $\forall n$ the measure of ω_n is bounded from $L|\Omega|$ so the bound pass to the limit.

We have the continuity for the solution of the Dirichlet problem, i.e if $\chi_{\omega_{n_k}} \to \chi_\omega$ in $L^1(\Omega)$ then $y_{\omega n_k} \to y_\omega$ in $L^2(\Omega)$. We can rewrite the system

$$\begin{cases} -\Delta y = \chi_{\omega_{n_k}} \\ y|_{\partial\Omega_{n_k}} = 0, \end{cases}$$

as

$$\int_\Omega |\nabla y_{\omega_{n_k}}|^2 dx = \int \chi_{\omega_{n_k}} y_{\omega_{n_k}} dx,$$

and we have that

$$\int \chi_{\omega_{n_k}} y_{\omega_{n_k}} dx \leq \|\chi_{\omega_{n_k}}\|_{L^2(\Omega)} \|y_{\omega_{n_k}}\|_{L^2(\Omega)} \leq C \|\chi_{\omega_{n_k}}\|_{L^1(\Omega)} \|y_{\omega_{n_k}}\|_{H_0^1(\Omega)},$$

where the last inequality follows by the Poincaré Inequality with $C > 0$ a constant.

For every $a, b, \varepsilon > 0 \in \mathbb{R}$ the following inequality holds

$$2ab \leq \varepsilon a^2 + \frac{1}{\varepsilon} b^2. \tag{3.4}$$

We apply the inequality above with $\varepsilon = \frac{1}{2}$ obtaining

$$C\|\chi_{\omega_{n_k}}\|_{L^2 1} \|y_{\omega_{n_k}}\|_{H_0^1(\Omega)} \leq \frac{C}{2}(2\|\chi_{\omega_{n_k}}\|^2_{L^1(\Omega)} + \frac{1}{2}\|y_{\omega_{n_k}}\|^2_{H_0^1(\Omega)}).$$

Since $\int_\Omega |\nabla y_{\omega_{n_k}}|^2 dx = \|y_{\omega_{n_k}}\|^2_{H_0^1(\Omega)}$ we have

$$(1 - \frac{C}{4})\|y_{\omega_{n_k}}\|^2_{H_0^1(\Omega)} \leq C\|\chi_{\omega_{n_k}}\|^2_{L^1(\Omega)},$$

and so

$$\|y_{\omega_{n_k}}\|^2_{H_0^1(\Omega)} \leq K, \tag{3.5}$$

with K a constant. The sequence $y_{\omega_{n_k}}$ is uniformly bounded in $H_0^1(\Omega)$ that is an Hilbert Space, so there exist an element y in $H_0^1(\Omega)$ such that the sequence weakly converges to y. This element is y_ω. We have that $\chi_{\omega_{n_k}} \to \chi_\omega$ in $L^1(\Omega)$, so it converges also in $L^2(\Omega)$ and also in the weak sense in $L^2(\Omega)$, therefore we have

$$\int \chi_{\omega_{n_k}} y_{\omega_{n_k}} dx \to \int \chi_\omega y_{\omega_{n_k}} dx.$$

From the variational formulation of the system we have that for every $\varphi \in H_0^1(\Omega)$,

$$\int_\Omega \nabla y_{\omega_{n_k}} \nabla \varphi dx = \int \chi_{\omega_{n_k}} \varphi,$$

passing to the limit in both sides we have that

$$\int_\Omega \nabla y \nabla \varphi dx = \int \chi_\omega \varphi.$$

This implies that $y = y_\omega$ and the weak convergence $y_{\omega_{n_k}} \overset{H_0^1(\Omega)}{\rightharpoonup} y_\omega$ is equivalent up to a subsequence to the strong convergence in $L^2(\Omega)$.

From proposition 1.14. the perimeter is a lower semicontinuous function and so is $\mathcal{J}$ being sum of two continuous function and a l.s.c. one. We have that $\left\|y_{\omega_{n_k}} - y_d\right\|^2_{L^2(\Omega)} \to \|y_\omega - y_d\|^2_{L^2(\Omega)}$ and $\left\|\chi_{\omega_{n_k}}\right\|_{L^1(\Omega)} \to \|\chi_\omega\|_{L^1(\Omega)}$, so the first two terms of the functional $\mathcal{J}$ are continuous. In the end it holds

$$\mathcal{J}(\omega) \leq \liminf(\left\|y_{\omega_{n_k}} - y_d\right\|^2_{L^2(\Omega)} + |\omega_{n_k}| + P_\Omega(\omega_{n_k})),$$

which proves that ω is the searched minimizer.

Now we consider the problem with the functional

$$\mathcal{J}(\omega) = \frac{1}{2} \|y_\omega - y_d\|^2_{L^2(\Omega)},$$

with $y_d \in L^2(\Omega)$ fixed, and with the set of possible shapes

$$\mathcal{U}_L = \{\omega \subset \Omega \ \mid \ |\omega| \leq L|\Omega| \quad \text{and} \quad \|\chi_\omega\|_{BV} \leq M\},$$

with $M > 0$ a fixed constant.

The functional $\mathcal{J}$ is obviously bounded from below from zero. Let ω_n be a minimizing sequence. By compactness of $BV(\Omega)$ we know that there exist a subsequence $\chi_{\omega_{n_k}}$ that converges to χ_ω in $L^1(\Omega)$. Reasoning as in the case above we have the continuity of the solution, we obtain that $y_{\omega_{n_k}}$ converges weakly in $H_0^1(\Omega)$ to y_ω, and strongly in $L^2(\Omega)$.

The considered functional $\mathcal{J}$ is continuous so we have that

$$\lim \mathcal{J}(\omega_{n_k}) = \mathcal{J}(\omega),$$

and ω is the minimizer.

To have the convergence of the limit it suffit to have a functional $J(\omega) = \int_\Omega j(x, r, p)$ where $r = y_\omega$ and $p = \nabla y_\omega$, such that exist a constant C with

$$|j(x, r, p)| \leq C(1 + r^2 + |p|^2)$$

.

3.2 Minimization of Coercive forms

First, we report some results regarding the minimization of functionals of a particular form.

Let $\mathcal{U}$ be a Hilbert space, the following data are given

- a continuous bi-linear form π on $\mathcal{U}$, which is symmetric $u, v \to \pi(u, v) = \pi(v, u)$,
- a continuous linear form L on $\mathcal{U}$, $v \to L(v)$,
- a closed, convex set $\mathcal{U}_{ad}$ in $\mathcal{U}$, called the set of admissible controls.

It is required to minimize the quadratic functional

$$J(v) = \pi(u, v) - 2L(v) \tag{3.6}$$

over the set $\mathcal{U}_{ad}$.

We consider a coercive form π on $\mathcal{U}$, i.e.

$$\pi(v, v) \geq c\|v\|^2 \qquad \forall v \in \mathcal{U},\ c > 0, \tag{3.7}$$

and we have the following theorem

Theorem 3.6. *Let $\pi(u, v)$ be a continuous symmetric bilinear coercive form on $\mathcal{U}$, and let J be of the form $J(v) = \pi(v, v) - 2L(v)$. Then there exists a unique element u in $\mathcal{U}_{ad}$ such that $J(u) = \inf_{v \in \mathcal{U}_{ad}} J(v)$.*

Proof. Let $v_n \in \mathcal{U}_{ad}$ be a minimizing sequence such that $J(v_n) = \inf_{v \in \mathcal{U}_{ad}} J(v)$. Since π is coercive and L is linear, we have

$$J(v) \geq c\|v\|^2 - k\|v\|, \qquad \text{where } c, c_1 \text{ are two constants.}$$

From above the sequence v_n is bounded, $\|v_n\| \leq K$, so we may extract a subsequence v_{n_k} such that

$$v_{n_k} \to w \qquad \textit{weakly in } \mathcal{U}.$$

Since $\mathcal{U}_{ad}$ is a closed convex set, it is weakly closed, it implies that $w \in \mathcal{U}_{ad}$.

But the function $v \to \pi(v, v)$ is lower semi-continuous (1.s.c.) in the weak topology of $\mathcal{U}$ and the function $L(v)$ is continuous in the weak topology. Thus the function $J(v)$ is weakly lower semi-continuous and hence $u = w$ is the searched minimizer.

We now prove the uniqueness, the function $v \to \pi(v, v)$ is strictly convex and the function $L(v)$ is linear. Hence the function $J(v)$ is also strictly convex.

We consider u_1 and u_2 two minimizer, since $\mathcal{U}_{ad}$ is a convex set the element $\frac{1}{2}(u_1 + u_2) \in \mathcal{U}_{ad}$ and we have $J(\frac{1}{2}(u_1 + u_2)) < J(u_1) = J(u_2)$, which is a contraddiction by supposing u_1, u_2 minimizes. □

Corollary 3.7. *Let $v \to J(v)$ be a convex functional from $\mathcal{U}_{ad}$ to $\mathbb{R}$, such that J is strongly lower semi-continuous and coercive, i.e.*

$$J(v) \to +\infty \qquad \|v\| \to +\infty.$$

Then there exists $u \in \mathcal{U}_{ad}$ that minimizes the functional.

We note that the minimizer is not necessarily unique, we have the uniqueness if we assume the functional J strictly convex.

We want to characterize the minimizing element and we have the following theorem

Theorem 3.8. *Let the assumptions of Theorem 2.1 remain valid. The minimizing element u of ad is characterized by*

$$\pi(u, v - u) \geq L(v - u), \qquad \forall v \in \mathcal{U}_{ad}. \tag{3.8}$$

Proof. First of all we prove that a minimizer u satisfies the condition (3.8). Let u be the minimizing element. Then for any $v \in \mathcal{U}_{ad}$ and $\theta \in (0, 1)$, we have

$$J(u) \leq J((1 - \theta)u + \theta v),$$

whence

$$\frac{1}{\theta}(J(u + \theta(v - u)) - J(u)) \geq 0.$$

Therefore, passing to the limit, we obtain,

$$\lim_{\theta \to 0} \frac{1}{\theta}(J(u + \theta(v - u)) - J(u)) \geq 0,$$

and if the limit exists, we may write the condition above as,

$$J'(u)(v - u) \geq 0. \tag{3.9}$$

Considering the functional $J(v) = \pi(v, v) - 2L(v)$, we have

$$\begin{aligned}
&\pi(u + \theta(v - u), u + \theta(v - u)) - 2L(u + \theta(v - u)) - \pi(u, u) + 2L(u) \\
&= \pi(u, u) + \theta^2\pi(v - u, v - u) + 2\theta\pi(u, v - u) - 2L(u) - 2\theta L(v - u) - \pi(u, u) + 2L(u) \\
&= \theta^2\pi(v - u, v - u) + 2\theta\pi(u, v - u) - 2L(u) - 2\theta L(v - u),
\end{aligned}$$

and the limit is of the form

$$\frac{1}{\theta}(\theta^2\pi(v-u,v-u)+2\theta\pi(u,v-u)-2L(u)-2\theta L(v-u)) = 2\pi(u,v-u)-2L(u)-2L(v-u).$$

So we can write that

$$J'(u)(v-u) = 2(\pi(u,v-u) - L(v-u)) \geq 0,$$

which proves the thesis. Alternatively now we suppose that (3.8) holds so u is a minimizer. The function $v \to J(v)$ is convex, we have for any $\theta \in (0,1)$,

$$J(v) - J(w) \geq \frac{1}{\theta}(J((1-\theta)w + \theta v)) - J(w)) \geq 0 \qquad \forall v, w \in \mathcal{U},$$

Passing to the limits as above

$$J(v) - J(w) \geq J'(w)(v-w), \tag{3.10}$$

and for $w = u$ by the (3.9) this is always positive. Hence

$$J(v) - J(u) \geq 0 \qquad \forall v \in \mathcal{U}_{ad}$$

which proves the theorem. □

We have the following remark in the case where $\mathcal{U}_{ad} = \mathcal{U}$

Corollary 3.9. *The condition* (3.8) *is equivalent to*

$$\pi(u,v) = L(v) \qquad \forall v \in \mathcal{U}. \tag{3.11}$$

The same proof works in the following setting, we have the theorem

Theorem 3.10. *Let $v \to J(v)$ be a strictly convex functional from $\mathcal{U}_{ad}$ to $\mathbb{R}$, differentiable and coercive, i.e.*

$$J(v) \to +\infty \qquad \|v\| \to +\infty.$$

Then there exist a unique element $u \in \mathcal{U}_{ad}$ that minimizes the functional, $J(u) = \inf_{v \in \mathcal{U}_{ad}} J(v)$, and is characterize by

$$J'(u)(v-u) \geq 0 \qquad \forall v \in \mathcal{U}_{ad}. \tag{3.12}$$

This last result could be rewritten using an equivalent characterization of the minimizer, we have the following result

Theorem 3.11. *Let all the hypotheses of Theorem 3.10 be satisfied. Then the characterization* (3.12) *is equivalent to*

$$J'(v)(v-u) \geq 0 \qquad \forall v \in \mathcal{U}_{ad} \tag{3.13}$$

To prove this result we need another theorem

Theorem 3.12. *Assume that the functional $v \to J(v)$ is a convex and differentiable functional from $\mathcal{U}_{ad}$ to $\mathbb{R}$. Then the derivative $v \to J'(v)$ has the following monotonicity property*

$$(J'(v) - J'(w))(v-w) \geq 0 \qquad \forall v, w \in \mathcal{U}_{ad}. \tag{3.14}$$

Proof. With this assumption the proof of the theorem 3.8. is still true and the condition (3.10) holds for every $v, w \in \mathcal{U}$. So from (3.10)we have that

$$J'(w)(v-w) \leq J(v) - J(w),$$

and interchanging v and w we have

$$J(w) - J(v) \geq J'(v)(w-v),$$

so we have

$$J'(v)(v-w) \leq J(v) - J(w).$$

Combining the two conditions above we have

$$(J'(v) - J'(w))(v-w) \geq J(v) - J(w) - J(v) + J(w) = 0.$$

□

Now we can write the proof of Theorem 3.11.

Proof. First we prove that condition (3.13) implies (3.12). In (3.13), we consider the function $v = (1-\theta)w + \theta u$ where w is any element of $\mathcal{U}_{ad}$ and $\theta \in (0,1)$, this implies

$$J'((1-\theta)w + \theta u)((1-\theta)(w-u)) = (1-\theta)J'((1-\theta)w + \theta u)(w-u) \geq 0,$$

and hence

$$J'((1-\theta)w + \theta u)(w-u) \geq 0.$$

If we let $\theta \to 1$, we then obtain

$$J'(u)(w-u) \geq 0 \qquad \forall w \in \mathcal{U}_{ad},$$

which is precisely (3.12). To prove the opposite implication we consider

$$J'(v)(v-u) = J'(u)(v-u) + (J'(v) - J'(u))(v-u).$$

From the result (3.14) we have that the second adding is positive therefore

$$J'(v)(v-u) \geq J'(u)(v-u) \geq 0,$$

so we have the thesis. □

So far we have assumed that $\mathcal{U}_{ad}$ is closed and convex. We have found that conditions (3.8), (3.12), (3.13) are necessary and sufficient to characterize the minimizer. Now we suppose $\mathcal{U}_{ad}$ is only closed and we consider the set defined by

$$\begin{aligned}\mathcal{C}(\mathcal{U}_{ad}; u) =& \{w \mid w \in \mathcal{U}_{ad}; \textit{there exist } u_n \in \mathcal{U}_{ad} \textit{ and } \lambda_n > 0, \\ & \textit{s.t. } u_n \to u \textit{ and } \lambda_n(u_n - u) \to w \textit{ in } \mathcal{U}\},\end{aligned}$$

that is a closed cone.

Theorem 3.13. *Let $v \to J(v)$ be a differentiable functional from $\mathcal{U}_{ad}$ to $\mathbb{R}$. We assume that exists an element $u \in \mathcal{U}_{ad}$ that minimizes the functional, $J(u) \leq J(v)$ $\forall v \in \mathcal{U}_{ad}$. Then*

$$J'(u)w \geq 0 \qquad \forall w \in \mathcal{C}(\mathcal{U}_{ad}; u) \tag{3.15}$$

Proof. Let u being the minimizing element and $w \in \mathcal{C}(\mathcal{U}_{ad}; u)$, utilizing the definition of the set $\mathcal{C}(\mathcal{U}_{ad}; u)$ we have

$$J(u) \leq J(u_n) = J(u + u_n - u) = J(u) + J'(u)(u_n - u) + o(\|u_n - u\|),$$

hence we have

$$J'(u)\lambda_n(u_n - u) + \lambda_n o(\|u_n - u\|) \geq 0.$$

Passing to the limit in n we obtain $\lambda_n(u_n - u) \to w$ and $\lambda_n o(\|u_n - u\|) = \lambda_n \|u_n - u\| o(1) \to 0$, so the (3.15) is proved. □

The condition (3.15) is a 'first order necessary condition'.

When the set $\mathcal{U}_{ad}$ is closed and convex then (3.15) is equivalent to (3.12). We first show that (3.15) implies (3.12). Let $v \in \mathcal{U}_{ad}$, since $\mathcal{U}_{ad}$ is convex $\forall \theta_n \in (0, 1)$

$$u_n = (1 - \theta_n)v + \theta_n u \in \mathcal{U}_{ad},$$

and hence

$$\frac{1}{1 - \theta_n}(u_n - u) = v - u.$$

Passing to the limit $\theta_n \to 1$ we have $u_n \to u$ and, called $\lambda_n = \dfrac{1}{1-\theta_n}$, we have that $\lambda_n(u_n - u) \to v - u$. So if we set $w = v - u$ we have $w \in \mathcal{C}(\mathcal{U}_{ad}, u)$, and the condition (3.15) implies (3.12).

Conversely, suppose (3.12) is true and we consider $w \in \mathcal{C}(\mathcal{U}_{ad}, u)$. For $u_n \in \mathcal{U}_{ad}$ s.t. $\lambda_n(u_n - u) \to w$ is true

$$J'(u)(u_n - u) \geq 0,$$

and hence $\lambda_n J'(u)(u_n - u) \geq 0$. Passing to the limit we obtain (3.15).

We now consider the (3.8), we want to give a prior characterization of the element u which satisfies that condition.

For a given bilinear form $\pi(u, v)$ on $\mathcal{U}$ and a given linear form L on $\mathcal{U}$, we have the theorem

Theorem 3.14. *Assume that the not necessarily symmetric bilinear form $\pi(u, v)$ satisfies*

$$\pi(v_1 - v_2, v_1 - v_2) \geq c\|v_1 - v_2\|^2 \qquad c > 0, \forall v_1, v_2 \in \mathcal{U}_{ad}. \tag{3.16}$$

Then there exists a unique u in $\mathcal{U}_{ad}$ which satisfies

$$\pi(u, v - u) \geq L(v - u) \qquad \forall v \in \mathcal{U}_{ad}.$$

In particular if $\mathcal{U}_{ad} = \mathcal{U}$, Theorem 3.14 yields the corollary

Corollary 3.15. *if $\pi(u, v)$ is a continuous bilinear form on $\mathcal{U}$ which is not necessarily symmetric and coercive, then there exists a unique $u \in \mathcal{U}$ such that*

$$\pi(u, v) = L(u) \qquad \forall v \in \mathcal{U}. \tag{3.17}$$

Let us consider the functional $v \to J(v)$ which is continuous, convex and lower semi-continuous in the weak topology. Let us denote by X the set of elements $u \in \mathcal{U}_{ad}$ such that

$$J(u) = \inf_{v \in \mathcal{U}_{ad}} J(v).$$

The set X may be empty. However if $\mathcal{U}_{ad}$ is also bounded then X is non-empty.

Theorem 3.16. *X is a closed convex subset of $\mathcal{U}_{ad}$.*

Proof. X is a closed subset as pre image of a closet set. To prove convexity, we have that if u_1, $u_2 \in X$

$$J(u_i) \leq J(v) \qquad \forall v \in \mathcal{U}_{ad},\ i = 1, 2.$$

If $\theta \in (0,1)$, we have that

$$J((1-\theta)u_1 + \theta u_2) \leq (1-\theta)J(u_1) + \theta J(u_2) \leq J(v) \qquad \forall v \in \mathcal{U}_{ad},$$

hence $(1-\theta)u_1 + \theta u_2 \in X$. □

We have the theorem

Theorem 3.17. *Let $\pi(u,v)$ be a continuous, not necessarily symmetric bilinear form on $\mathcal{U}$ satisfying*

$$\pi(v,v) \geq 0 \qquad \forall v \in \mathcal{U}_{ad}.$$

Then the set X of solutions of

$$\pi(u, v-u) \geq L(v-u) \qquad \forall v \in \mathcal{U}_{ad} \tag{3.18}$$

is closed, convex subset of $\mathcal{U}$.

Proof. From theorem 3.11. we have that (3.18) is equivalent to

$$\pi(v, v-u) \geq L(v-u) \qquad \forall v \in \mathcal{U}_{ad}.$$

If we consider u_1 and u_2 in X, for $\theta \in (0,1)$ we have that

$$\begin{aligned}
&\pi(v, v-((1-\theta)u_1 + \theta u_2)) = \pi(v, (1-\theta)(v-u_1) + \theta(v-u_2) \\
&= (1-\theta)\pi(v,u_1) + \theta\pi(v,u_2) \geq (1-\theta)L(v-u_1) + \theta L(v-u_2) \\
&\leq L(v-(1-\theta)u_1 + \theta u_2),
\end{aligned}$$

so $(1-\theta)u_1 + \theta u_2 \in X$. □

3.3 Control of Elliptic Variational Problems

Let V and H be two Hilbert Spaces on $\mathbb{R}$. Assume $V \subseteq H$, the injection of V into H is continuous, and V is dense in H. H will be identified with its dual space. If V' denotes the dual space of V, H may be identified with a subspace of V' and we may write

$$V \subseteq H \subseteq V'. \tag{3.19}$$

The following data are given:

- let $a(u,v)$ be a continuous, coercive, bilinear form on V, that is

$$a(v,v) \geq \alpha \|v\|^2, \ \alpha > 0, \tag{3.20}$$

- let L be a continuous linear form on V so it can be seen as a scalar product between $f \in V'$ and $v \in V$,

$$L(v) = (f, v). \tag{3.21}$$

The scalar product $(,) = (,)_{V',V}$ coincides with the scalar product in H if $f \in H$.

We identify the Hilbert space $V = \mathcal{U}$, and the bilinear form $a(u, v) = \pi(u, v)$. From the (3.17) we know that there exists a unique $u \in \mathcal{U}$ such that

$$a(u, v) = L(v) = (f, v) \qquad \forall v \in \mathcal{U}.$$

We can rewrite this by

Theorem 3.18. *Let $a(u, v)$ be a bilinear form satisfying the conditions* (3.20) *and let $f \in V'$, there exists a unique $y \in V$ such that*

$$a(y, \psi) = (f, \psi) \quad \forall \psi \in V. \tag{3.22}$$

Proof. We apply the corollary 3.15 with the bilinear form $a(u, v)$ that satisfies the conditions in (3.20) and the linear form L which satisfies the condition (3.21). □

By the Lax-Milgram theorem we have that exist a linear operator $A \in \mathcal{L}(V, V')$ such that

$$a(u, \psi) = (Au, \psi) \qquad Au \in V'. \tag{3.23}$$

If we fix the element y we may interpret the bilinear form a as a linear form $\psi \to a(y, \psi)$ that belongs to V'. We define the linear operator A so that $Ay = v_y \in V'$ such that for every $\psi \in V$ the evaluation of v_y in ψ coincides with $a(y, \psi)$. Hence from (3.22) this is equivalent to

$$Ay = f. \tag{3.24}$$

Formulation of the first control problem

We can now formulate the first control problem. Let a Hilbert space $\mathcal{U}$ be the space of controls and let $B \in \mathcal{L}(\mathcal{U}, V')$ be a given operator. A system governed by the operator A as in (3.24) is given. For a control $u \in \mathcal{U}$, we define the *state equation* associated to the system by

$$Ay = f + Bu, \tag{3.25}$$

whose solution $y \in V$ is called *state*.

The state variable y depends on u so we can rewrite the equation as

$$Ay(u) = f + Bu, \qquad y(u) \in V. \tag{3.26}$$

We also define the *observation equation* as

$$z(u) = Cy(u), \tag{3.27}$$

where $C \in \mathcal{L}(V, \mathcal{H})$, with $\mathcal{H}$ an Hilbert Space.

Finally we are given

$$\begin{aligned} N \in \mathcal{L}(\mathcal{U},\mathcal{U}) \qquad & N \text{ hermitian, definite positive,} \\ & (Nu, u) \geq c\|u\|^2, c > 0. \end{aligned} \tag{3.28}$$

With every control $u \in \mathcal{U}$ we associate the cost functional

$$J(u) = \|Cy(u) - z_d\|^2_{\mathcal{H}} + (Nu, u)_{\mathcal{U}}, \tag{3.29}$$

where z_d is a given element in $\mathcal{H}$.

Let $\mathcal{U}_{ad}$ be a closed convex subset of $\mathcal{U}$. The control problem consists of finding the

$$\inf_{v \in \mathcal{U}_{ad}} J(v).$$

The map that associate $u \to y(u)$ is linear and C is a linear operator, therefore we can write the functional J in the form

$$J(u) = \|C(y(u) - y(0)) + Cy(0) - z_d\|^2_{\mathcal{H}} + (Nu, u)_{\mathcal{U}}. \tag{3.30}$$

We set

$$\pi(u, v) = (C(y(u) - y(0)), C(y(v) - y(0)))_{\mathcal{H}} + (Nu, v)_{\mathcal{U}}, \tag{3.31}$$

and

$$L(v) = (z_d - Cy(0), C(y(v) - y(0)))_{\mathcal{H}}. \tag{3.32}$$

Since the map $u \to y(u)$ is linear and C is a linear operator, $L(v)$ is a linear and continuous form on $\mathcal{U}$ and $\pi(u, v)$ is a continuous bilinear form on $\mathcal{U}$.

We can rewrite the functional as

$$J(v) = \pi(v, v) - 2L(v) + \|z_d - Cy(0)\|^2_{\mathcal{H}}. \tag{3.33}$$

This follows from

$$\begin{aligned}
J(v) &= (C(y(v)-y(0))+Cy(0)-z_d, C(y(v)-y(0))+Cy(0)-z_d)_{\mathcal{H}} + (Nv,v)_{\mathcal{U}} \\
&= (C(y(v)-y(0)), C(y(v)-y(0)))_{\mathcal{H}} + (C(y(v)-y(0)), Cy(0)-z_d)_{\mathcal{H}} \\
&+ (Cy(0)-z_d, C(y(v)-y(0)))_{\mathcal{H}} + (Cy(0)-z_d, Cy(0)-z_d)_{\mathcal{H}} + (Nv,v)_{\mathcal{U}} \\
&= (C(y(v)-y(0)), C(y(v)-y(0)))_{\mathcal{H}} + (Nv,v)_{\mathcal{U}} \\
&- 2(z_d - Cy(0), C(y(v)-y(0)))_{\mathcal{H}} + (z_d - Cy(0), z_d - Cy(0))_{\mathcal{H}} \\
&= \pi(v,v) - 2L(v) + \|z_d - Cy(0)\|^2_{\mathcal{H}}.
\end{aligned}$$

Moreover from (3.28) and since $(C(y(u)-y(0)), C(y(u)-y(0)))_{\mathcal{H}} \geq 0$ we have that

$$\pi(u,u) \geq c\|u\|^2 \qquad u \in \mathcal{U}. \tag{3.34}$$

Since $\|z_d - Cy(0)\|^2_{\mathcal{H}} \geq 0$ we have reduced the problem to a form where theorem 3.6. can be applied.

Thus we have

Theorem 3.19. *Let us suppose that conditions* (3.34) *and* (3.28) *hold and that the state of the system is given by* (3.25)*. Then there exist a unique element* $u \in \mathcal{U}_{ad}$ *such that*

$$J(u) = \inf_{v \in \mathcal{U}_{ad}} J(v).$$

The element $u \in \mathcal{U}_{ad}$ for which $J(v)$ attains the minimum is termed the *optimal control*.

The case when $N = 0$, we are only able to conclude that $\pi(v,v) \geq 0$, thus we have the theorem

Theorem 3.20. *Let us suppose that* $\pi(u,v)$ *satisfies condition* (3.20) *hold and* $N = 0$*. We also assume that* $\mathcal{U}_{ad}$ *is bounded. Then there exists a non empty subset* X *of* $\mathcal{U}_{ad}$ *such that*

$$J(u) = \inf_{v \in \mathcal{U}_{ad}} J(v) \qquad \forall u \in X.$$

The set X *is closed and convex.*

Proof. It follows by the application of the theorem 3.16. □

The set X is termed the set of optimal controls.

So the existence of at least a minimizer follows by the theory of the previous section regarding the coercive form, the difference lies in the fact that now we don't have a functional $\mathcal{J}$ depending directly from the control, but we need two operators

- the operator $u \to y(u)$ which transforms the control u into a state $y(u)$,
- the operator $y(u) \to Cy(u)$ which transforms the state into an observation.

Now we want to write down a set of equations defining the optimal control to characterize it.

3.3.1 Optimality Condition

If u is an optimal control, from theorem 3.11. it satisfies (3.12)

$$J'(u)(v-u) \geq 0 \qquad \forall v \in \mathcal{U}_{ad}, \tag{3.35}$$

and conversely. We can apply it because we have written the functional J in the form (3.33), so the condition (3.12) is the same as (3.9) of theorem 3.8., and J is convex, differentiable and tend to $+\infty$ when $\|u\| \to +\infty$.

Since A is an isomorphism from V onto V', we may write

$$y(u) = A^{-1}(f + Bu),$$

then

$$y'(u) \cdot \psi = A^{-1}B\psi,$$

hence

$$\begin{aligned} y'(u) \cdot (v-u) = A^{-1}B(v-u) &= A^{-1}B(v-u) + A^{-1}f - A^{-1}f \\ &= A^{-1}(f+Bv) - A^{-1}(f+Bu) = y(v) - y(u). \end{aligned}$$

Therefore (3.35) is equivalent to

$$(Cy(u) - z_d, C(y(v) - y(u)))_{\mathcal{H}} + (Nu, v-u)_{\mathcal{U}} \geq 0 \qquad \forall v \in \mathcal{U}_{ad}. \tag{3.36}$$

Let us introduce the dual $\mathcal{H}'$ of $\mathcal{H}$ and set

$$\Lambda = \textit{ canonical isomorphism of } \mathcal{H} \textit{ into } \mathcal{H}'.$$

Then we denote with $C^* \in \mathcal{L}(\mathcal{H}', V')$ the adjoint of C, for φ and ψ in V, we have

$$(C^*\Lambda C\psi, \varphi)_{V',V} = (\Lambda C\psi, C\varphi)_{\mathcal{H}',\mathcal{H}} = (C\psi, C\varphi)_{\mathcal{H}}. \tag{3.37}$$

Therefore (3.36) is equivalent to

$$(C^*\Lambda(Cy(u) - z_d), y(v) - y(u))_{V',V} + (Nu, v-u)_{\mathcal{U}} \geq 0 \quad \forall v \in \mathcal{U}_{ad}. \tag{3.38}$$

Let $A^* \in \mathcal{L}(V', V)$ be the adjoint of A, it is associated with a bilinear form

$$(A^*\varphi, \psi) = (\varphi, A\psi) = a(\psi, \varphi) \qquad \varphi, \psi \in V, \tag{3.39}$$

and A^* is an isomorphism between V and V'.

For a control $v \in \mathcal{U}$, we define the *adjoint state* $p(v) \in V$ defined by:

$$A^*p(v) = C^*\Lambda(Cy(v) - z_d). \tag{3.40}$$

We may rewrite (3.38) as

$$\begin{aligned}(C^*\Lambda(Cy(u) - z_d), y(v) - y(u)) &= (A^*p(u), y(v) - y(u)) \\ = (p(u), Ay(v) - Ay(u)) &= (p(u), B(v - u)) = (B^*p(u), v - u),\end{aligned} \tag{3.41}$$

where $B^* \in \mathcal{L}(V,\mathcal{U}')$ is the adjoint of B and the third equality follows from (3.26).

Whence (3.38) is equivalent to

$$(B^*p(u), v - u)_{\mathcal{U}',\mathcal{U}} + (Nu, v - u)_{\mathcal{U}} \geq 0 \quad \forall v \in \mathcal{U}_{ad}. \tag{3.42}$$

We now set the canonical isomorphism between $\mathcal{U}$ and $\mathcal{U}'$

$$\Phi = \textit{ canonical isomorphism of } \mathcal{U} \textit{ into } \mathcal{U}'.$$

We have for $v \in \mathcal{U}$, $\psi \in V$

$$(\psi, Bv)_{V,V'} = (B^*\psi, v)_{\mathcal{U}',\mathcal{U}} = (\Phi^{-1}B^*\psi, v)_{\mathcal{U}}.$$

Hence the (3.38) is equivalent to (3.42) that is equivalent to

$$(\Phi^{-1}B^*p(u) + Nu, v - u)_{\mathcal{U}} \geq 0 \qquad \forall v \in \mathcal{U}_{ad}. \tag{3.43}$$

We can summarize the results in the following theorem

Theorem 3.21. *Assume that* (3.20) *holds. The cost function being given by* (3.29)*, a necessary and sufficient condition for* u *to be an optimal control is that the following equations and inequalities be satisfied:*

1) $Ay(u) = f + Bu,$ (3.25)

2) $A^*p(v) = C^*\Lambda(Cy(v) - z_d),$ (3.40)

3) $(\Phi^{-1}B^*p(u) + Nu, v - u)_{\mathcal{U}} \geq 0 \qquad \forall v \in \mathcal{U}_{ad}.$ (3.43)

If N satisfies (3.28)*, then the optimal control is unique.*

If $N = 0$ and $\mathcal{U}_{ad}$ is bounded then the system of (3.25), (3.40), (3.43) *admits at least one solution. The set of solutions corresponds to a family of optimal controls forming a closed, convex subset X of $\mathcal{U}_{ad}$.*

We observe that (3.43) may be written in the equivalent form

$$(\Phi^{-1}B^*p(u) + Nu, u)_{\mathcal{U}} = \inf_{v\in\mathcal{U}_{ad}} (\Phi^{-1}B^*p(u) + Nu, v)_{\mathcal{U}}. \tag{3.44}$$

This last Theorem 3.21. is the Pontryagin's minimum principle, we can reason in the same way defining the adjoint state with the opposite sign

$$-A^*p(v) = C^*\Lambda(Cy(v) - z_d). \tag{3.45}$$

Thus we have that

$$(C^*\Lambda(Cy(u) - z_d), y(v) - y(u)) = (-A^*p(u), y(v) - y(u)) = (A^*p(u), y(u) - y(v))$$
$$= (p(u), Ay(u) - Ay(v)) = (p(u), B(u - v)) = (B^*p(u), u - v),$$

thus (3.38) is equivalent to

$$(B^*p(u), u - v)_{\mathcal{U}',\mathcal{U}} - (Nu, u - v)_{\mathcal{U}} \geq 0 \quad \forall v \in \mathcal{U}_{ad},$$

and we have that

$$(\Phi^{-1}B^*p(u) - Nu, u - v)_{\mathcal{U}} \geq 0 \qquad \forall v \in \mathcal{U}_{ad}. \tag{3.46}$$

This is the Pontryagin's maximum principle, in fact we have that the optimal control u is

$$(\Phi^{-1}B^*p(u) - Nu, u)_{\mathcal{U}} = \sup_{v\in\mathcal{U}_{ad}} (\Phi^{-1}B^*p(u) - Nu, v)_{\mathcal{U}}. \tag{3.47}$$

Now we want to give a result in a case where $N = 0$ but we can still deduce the unicity of the minimizer.

We consider the Hilbert spaces $\mathcal{U} = V'$ and $\mathcal{H} = V$. Thus we have that

$$B = \text{Identity map} \qquad C = \text{Identity map}$$

$$\Lambda = \text{canonical isomorphism of } V \text{ in } V'$$

$A \in \mathcal{L}(V, V')$ that define the state equation

$$Ay(u) = f + u, \tag{3.48}$$

and the observation is $z(u) = y(u)$.

The cost functional $J(v)$ is given by

$$J(v) = \|y(v) - z_d\|_V^2,$$

with z_d a given element in V.

We have the following theorem

Theorem 3.22. *Under the hypothesis above and hypotesis* (3.20) *being satisfied. There exist a unique optimal control such*

$$\mathcal{J}(u) = \inf_{v \in \mathcal{U}_{ad}} J(v).$$

Then the element u is given by the solution of

$$Ay(u) = f + u, \tag{3.49}$$

$$A^* p(v) = \Lambda(y(v) - z_d), \tag{3.50}$$

$$(p(u), v - u)_{V,V'} \geq 0 \qquad \forall v \in \mathcal{U}_{ad} \tag{3.51}$$

Proof. In this case (3.31) is

$$\pi(u, v) = ((y(u) - y(0)), (y(v) - y(0)))_V = (A^{-1}u, A^{-1}v)_V,$$

and therefore

$$\pi(v, v) = \|A^{-1}v\|_V^2 \geq c\|v\|_{V'}^2,$$

because A is an isomorphism so it suffit a $c < 1$. Hence we can apply Theorem 3.19. and Theorem 3.21. to conclude. □

Chapter 4

The study of a specific Shape Optimization Problem

For a fixed open bounded set Ω in $\mathbb{R}$, we consider a minimization problem with the set of possible shapes

$$\mathcal{U}_L = \{\omega \subset \Omega \ \mid \ |\omega| \leq L|\Omega|\},$$

a state equation of the form

$$\begin{cases} -\Delta y = \chi_\omega \\ y|_{\partial\Omega} = 0, \end{cases}$$

and a functional $\mathcal{J}$

$$\mathcal{J}(\omega) = \frac{1}{2} \left\| y_\omega - y_d \right\|^2_{L^2(\Omega)}.$$

We want to apply the technique of relaxation to this problem.

4.1 Relaxation of the problem

First of all we want to widen the set of the admissible shapes to make it compact with respect to a certain type of convergence. We identify Ω and every subset ω with their respective characteristic functions. The set $\mathcal{U}_L$ can be described as the following subset of $L^\infty(\Omega)$

$$\mathcal{U}_L = \{\ell \in L^\infty(\Omega) \ \mid \ 0 \leq \ell \leq 1, \ \ell(1-\ell) = 0 \ a.e., \ \int_\Omega \ell(x)dx \leq L|\Omega|\}.$$

We consider his closure in the weak-* topology of $L^\infty(\Omega)$.

Proposition 4.1. *The closure of the set $\mathcal{U}_L$ of characteristic functions, for the weak-* topology of L^∞ is*

$$\overline{\mathcal{U}_L} = \{a \in L^\infty(\Omega) \ \mid \ 0 \leq a \leq 1, \ \int_\Omega a(x)dx \leq L|\Omega|\}. \tag{4.1}$$

In other words $\overline{\mathcal{U}_L}$ is the weak- closed convex hull of $\mathcal{U}_L$ in $L^\infty(\Omega)$ and $\mathcal{U}_L$ is the set of extreme points of the convex set $\overline{\mathcal{U}_L}$.*

Now we write the relaxed state equation as

$$\begin{cases} -\Delta y = a \\ y|_{\partial\Omega} = 0, \end{cases} \tag{4.2}$$

and call y_a its unique solution in $H_0^1(\Omega)$. we search $a \in \overline{\mathcal{U}_L}$ that minimizes the functional $\mathcal{J}(a)$.

Theorem 4.2. *The relaxed problem* (4.2) *admits a unique minimizer.*

Proof. The functional $\mathcal{J}$ is precisely of the form squared norm of the Hilbert $H_0^1(\Omega)$, so we can apply theorem 3.19 and theorem 3.22 to conclude that a unique optimal solution for the relaxed problem always exists. □

We could also apply the classic method of Calculus of Variations to find the solution. The set $\overline{\mathcal{U}_L}$ is compact for the weak*-convergence in $L^\infty(\Omega)$ and the functional $\mathcal{J}$ is continuous. It remains to demonstrate that we have continuity for the solution y_a with respect to the weak*-convergence in $L^\infty(\Omega)$ of the control a.

For the same reasoning as the examples in chapter 3., we prove that the functions y_a are uniformly bounded.

We can rewrite the system (4.2) in the following variational form

$$\int_\Omega |\nabla y_a|^2 dx = \int a \ y_a dx.$$

The function $a \in \overline{\mathcal{U}_L}$ belong to $L^p(\Omega)$ for every $p \in [1, +\infty]$ because they are bounded above by 1 and Ω is a bounded set in $\mathbb{R}^n$ so it has finite Lebesgue measure. We have

$$\int a \ y_a dx \leq \|a\|_{L^2(\Omega)} \|y_a\|_{L^2(\Omega)} \leq C\|a\|_{L^2(\Omega)} \|y_a\|_{H_0^1(\Omega)},$$

where the last inequality follows by the Poincaré Inequality with $C > 0$ a constant.

Applying the inequality (3.4) above with $\varepsilon = \frac{1}{2}$ we obtain

$$C\|a\|_{L^2(\Omega)}\|y_a\|_{H_0^1(\Omega)} \leq \frac{C}{2}(2\|a\|^2_{L^2(\Omega)} + \frac{1}{2}\|y_a\|^2_{H_0^1(\Omega)}).$$

Whence we have

$$(1 - \frac{C}{4})\|y_a\|^2_{H_0^1(\Omega)} \leq 2C\|a\|^2_{L^1(\Omega)},$$

and

$$\|y_a\|^2_{H_0^1(\Omega)} \leq \tilde{C}\|a\|^2_{L^2(\Omega)}. \tag{4.3}$$

We consider a minimizing sequence $a_n \overset{*}{\rightharpoonup} a$, if we call y_{a_n} the solution of the system (4.2) obtained replacing a with a_n, we state that $y_{a_n} \to y_a$ in $L^2(\Omega)$. The sequence y_{a_n} is uniformly bounded from $sup_n\|a_n\|^2_{L^2(\Omega)} \leq L|\Omega| = K$, i.e.

$$\|y_{a_n}\|^2_{H_0^1(\Omega)} \leq K \qquad \forall n, \tag{4.4}$$

since $H_0^1(\Omega)$ is an Hilbert Space, there exists an element y in $H_0^1(\Omega)$ such that the sequence weakly converges to y. This element is y_a. The solution y_{a_n} are also in $L^1(\Omega)$, therefore we have

$$\int a_n \; y_{a_n} dx \to \int a \; y_{a_n} dx.$$

From the variational formulation of the system, we have that for every $\varphi \in C_c^\infty(\Omega)$,

$$\int_\Omega \nabla y_{a_n} \nabla \varphi dx = \int a_n \; \varphi \; dx,$$

passing to the limit in both sides ($\varphi \in C_c^\infty(\Omega)$ and Ω is bounded hence it belongs also in $L^1(\Omega)$) we have that

$$\int_\Omega \nabla y \nabla \varphi dx = \int a \; \varphi \; dx.$$

Since the last equality is valid for all $\varphi \in C_c^\infty(\Omega)$ we deduce by density of $C_c^\infty(\Omega)$ in $H_0^1(\Omega)$ that it holds for every function in $H_0^1(\Omega)$. This implies that $y = y_a$ and the weak convergence $y_{a_n} \overset{H_0^1(\Omega)}{\rightharpoonup} y_a$ is equivalent up to a subsequence to the strong convergence in $L^2(\Omega)$.

4.1.1 Optimality conditions

We apply the theory of chapter 3. to our problem.

We consider the Hilbert spaces

$$\mathcal{U} = \mathcal{H} = L^2(\Omega) \qquad V = H_0^1(\Omega),$$

and

$$V' = H_0^1(\Omega)' = H^{-1}(\Omega) \supseteq L^2(\Omega).$$

Since

$$\mathcal{U}' = \mathcal{U} = \mathcal{H}' = \mathcal{H} = L^2(\Omega),$$

we have that the two canonical isomorphism Λ and Ψ are the identity map.

The set $\overline{\mathcal{U}_L}$ is contained in $L^\infty(\Omega) \subseteq L^2(\Omega)$, so we have $\overline{\mathcal{U}_L} \subseteq \mathcal{U}$.

We set the operator A,B and C as

$$A = -\Delta : H_0^1(\Omega) \to H^{-1}(\Omega),$$

with a state equation

$$-\Delta y(a) = a,$$

thus $f = 0$ in this case. We define the operators

$$B : L^2(\Omega) \to H^{-1}(\Omega) \text{ the continuous embedding of } L^2(\Omega) \text{ in } H^{-1}(\Omega),$$

$$C : H_0^1(\Omega) \to L^2(\Omega) \text{ the continuous embedding of } H_0^1(\Omega) \text{ in } L^2(\Omega).$$

The functional $\mathcal{J} = \frac{1}{2} \|y_\omega - y_d\|^2_{L^2(\Omega)}$ is of the form $\|Cy(u) - z_d\|^2$, with $\pi(v, v)$ that satisfies condition (3.34).

We can apply Theorem 3.21., with the formulation of the maximum principle, to deduce the following optimality conditions. If $\bar{a}$ is the optimal control and $\bar{p}$ is the adjoint state we have that

1) The first condition $Ay(u) = f + Bu$ becomes $-\Delta y(a) = a$,

2) $-A^* p(v) = C^* \Lambda (Cy(v) - z_d)$ becomes $\Delta p(a) = y(a) - y_d$,

3) $(\Phi^{-1} B^* p(u) - Nu, u - v)_{\mathcal{U}} \geq 0 \quad \forall v \in \mathcal{U}_{ad}$ becomes $(p(a), \bar{a} - a)_{L^2(\Omega)} \geq 0$.

If $\bar{a}$ is the optimal control and $\bar{p}$ is the adjoint state we summarize it in

$$\begin{cases} -\Delta y_{\bar{a}} = \bar{a}, & y_{\bar{a}}|_{\partial\Omega} = 0 \\ \Delta \bar{p} = y_{\bar{a}} - y_d, & \bar{p} = 0 \ in \ \partial\Omega \\ \int_\Omega \bar{p} \cdot (\bar{a} - a) dx \geq 0 & \forall a \in \overline{\mathcal{U}_L} \end{cases} \tag{4.5}$$

We consider now the Bathtub principle

Theorem 4.3. *Let (Ω, Σ, μ) be a measure space and let f be a real valued function on Ω such that $\mu\{x : f(x) < t\}$ is finite for all $t \in \mathbb{R}$. Let the number $G > 0$ be given and define a class of measurable functions in Ω by*

$$C = \{g : \ 0 \leq g(x) \leq 1 \text{ for all } x \text{ and } \int_\Omega g(x)\mu(dx) = G\}$$

Then the minimization problem $I = \inf_{g \in C} \int f(x)g(x)\mu(dx)$ is solved by

$$g(x) = \chi_{\{f<s\}}(x) + c\chi_{\{f=s\}}(x),$$

where $s = \sup\{t : \mu\{x : f(x) < t\} \leq G)\}$ and $c\mu\{(x : f(x) = s)\} = G - \mu\{x : f(x) < s\}$.

Proof. Let h be any other member of C. To show that g is a minimizer, you need to establish that $I(g) \leq I(h)$, which is equivalent to showing

$$\int_\Omega f(g - h) \leq 0.$$

To show this, split the range of integration into the sub-level, sup-level, and level sets of f, i.e. $\{f < s\}, \{f > s\}$ and $\{f = s\}$.

$$\begin{aligned}
\int_\Omega f(g-h)d\mu &= \int_{\{f<s\}} f(g-h)d\mu + \int_{\{f>s\}} f(g-h)d\mu + \int_{\{f=s\}} f(g-h)d\mu \\
&\leq s\int_{\{f<s\}} (g-h)d\mu - \int_{\{f>s\}} fhd\mu + \int_{\{f=s\}} s(g-h)d\mu \\
&\leq s\int_{\{f<s\}} (g-h)d\mu - s\int_{\{f>s\}} hd\mu + s\int_{\{f=s\}} (g-h)d\mu \\
&\leq s(\int_{\{f<s\}} (g-h)d\mu + \int_{\{f>s\}} (g-h)d\mu + \int_{\{f=s\}} (g-h)d\mu) \\
&= s\int_\Omega (g-h)d\mu = s(G-G) = 0,
\end{aligned}$$

where the inequality at the fourth passage is true because we know that $g = 0$ in $\{f < s\}$. □

We apply this theorem to our problem, we consider $C = \overline{\mathcal{U}_L}$ with $\mu(dx)$ the Lebesgue measure and $f = -\bar{p}$. From the third optimality condition, we have

$$\int_\Omega \bar{p}\ \bar{a}\ dx \geq \int_\Omega \bar{p}\ a\ dx \qquad \forall a \in \overline{\mathcal{U}_L},$$

so

$$I = \sup_{a \in \overline{\mathcal{U}_L}} \int_\Omega \bar{p}\ a\ dx$$

is solved by $\bar{a}$.

For the Bathtub principle above $\bar{a}$ is of the form

$$\bar{a} = \chi_{\{\bar{p} > s\}} + c\chi_{\{\bar{p}=s\}}, \tag{4.6}$$

where s is defined by

$$s = \inf\{\sigma \in \mathbb{R} \ \mid \ |\{\bar{p} > \sigma\}| \leq L|\Omega|\},$$

and $c \in L^\infty(\Omega; [0,1])$.

We can write the following theorem

Theorem 4.4. *Let Ω be an open bounded set of $\mathbb{R}^n$ with the Lebesgue measure $\mathcal{L}^n$ and let $\bar{p}$ be the adjoint state of the problem. Let $\overline{\mathcal{U}_L}$ be the set defined in* (4.1)*. Then the maximization problem $I = \sup_{a \in \overline{\mathcal{U}_L}} \int \bar{p}(x)a(x)dx$ is solved by*

$$\bar{a} = \chi_{\{\bar{p} > s\}} + c\chi_{\{\bar{p}=s\}}, \tag{4.7}$$

where $s = \inf\{\sigma \in \mathbb{R} \ \mid \ |\{\bar{p} > \sigma\}| \leq L|\Omega|\}$ and $c \in L^\infty(\Omega; [0,1])$.

Proof. We consider for every $G \leq L|\Omega|$ the level of the set $\overline{\mathcal{U}_L}$

$$\overline{\mathcal{U}_G} = \{a \in L^\infty(\Omega) \ \mid \ 0 \leq a \leq 1, \ \int_\Omega a(x)dx = G\}.$$

For every G we consider the problem $I_G = \sup_{a \in \overline{\mathcal{U}_G}} \int \bar{p}(x)a(x)dx$.

If we consider $f = -\bar{p}$, we have that $I_G = \inf_{a \in \overline{\mathcal{U}_G}} \int f(x)a(x)dx$ We know from the Bathtub principle that the minimizer is the function

$$a_G(x) = \chi_{\{f > s_G\}} + c_G\chi_{\{f=s_G\}},$$

with $s_G = \sup\{t : |\{f(x) < t\}| \leq G)\}$ and $c_G = \dfrac{G - |\{f(x) < s_G\}|}{|\{f(x) = s_G)\}|}$.

Substituting $f = -\bar{p}$ we have

$$a_G(x) = \chi_{\{-\bar{p} < s_G\}} + c_G\chi_{\{-\bar{p}=s_G\}},$$

$s_G = \sup\{t : |\{-\bar{p} < t\}| \leq G)\}$ and $c_G = \dfrac{G - |\{\bar{p} < s_G\}|}{|\{\bar{p} = s_G\}|}$.

Then we have that $\{-\bar{p} < t\} = \{\bar{p} > -t\}$ and we have that $s_G = \sup\{t : |\{\bar{p} > -t\}| \leq G)\} = \inf\{t : |\{\bar{p} > t\}| \leq G)\}$.

For $G_1 > G_2$ we have that $s_{G_2} \geq s_{G_1}$ because the set $\{t : |\{\bar{p} > t\}| \leq G_2)\} \subseteq \{t : |\{\bar{p}(x) > t\}| \leq G_1)\}$. Thus we have that the set $\{\bar{p} > s_{G_1}\} \subseteq \{\bar{p} < s_{G_2}\}$.

We want to minimize over the set $\overline{\mathcal{U}_L} = \bigcup_{G \leq L|\Omega|} \overline{\mathcal{U}_G}$, so the value s associated to the optimal control is

$$s = \inf \bigcup_{G \leq L|\Omega|} \{t : |\{\bar{p} > t\}| \leq G)\} = \sup\{t : |\{\bar{p} > t\}| \leq L|\Omega|)\} = s_{L|\Omega|}.$$

For all $G \leq L|\Omega|$ we have $s_G \leq s$, hence the level set $\{\bar{p} = s_G\}$ could be contained in the set $\{\bar{p} < s\}$ or in the set $\{\bar{p} = s\}$. For every G we have an associated constant c_G, so the function c associated to the level set $\{\bar{p} = s\}$ when we minimize over all the possible set $\mathcal{U}_G$ is a function that could assume every possible value c_G. In particular $c_G \in [0, 1]$ for every G so we can write $c \in L^\infty(\Omega, [0, 1])$. □

We remark that we could characterize the minimum reasoning the same way also using Theorem 3.21. with the minimum principle. In this case the adjoint state is

$$-\Delta\bar{p} = y_{\bar{a}} - y_d, \qquad \bar{p} = 0 \ in \ \partial\Omega,$$

and the third optimality condition is written in the form

$$\int_\Omega \bar{p} \cdot (a - \bar{a}) dx \geq 0 \qquad \forall a \in \overline{\mathcal{U}_L}.$$

Thus the optimal control $\bar{a}$ solves the problem

$$\int_\Omega \bar{p}\ \bar{a} = \inf_{a \in \overline{\mathcal{U}_L}} \int_\Omega \bar{p}\ a\ dx,$$

and applying the Bathtub principle we have $\bar{a}$ is of the form

$$\bar{a} = \chi_{\{\bar{p} < s\}} + c\chi_{\{\bar{p} = s\}}, \tag{4.8}$$

where s is defined by

$$s = \sup\{\sigma \in \mathbb{R} \ \mid \ |\{\bar{p} < \sigma\}| \leq L|\Omega|\},$$

and $c \in L^\infty(\Omega; [0, 1])$. This other formulation is equivalent to the formulation above.

4.1.2 Existence of optimal shapes

We are interested in the case where the optimal solution of the relaxed problem is indeed a shape, so a characteristic function. In some cases, giving a constraint on y_d, we are able to prove the existence of a minimizer for the original problem.

We consider the solution y^0 of the system with $a = 0$

$$\begin{cases} -\Delta y^0 = 0 \quad in\ \Omega \\ y|_{\partial\Omega} = 0, \end{cases} \tag{4.9}$$

that is indeed $y^0 = 0$, and the solution y^1 of the system with $a = 1$

$$\begin{cases} -\Delta y^1 = 1 \quad in\ \Omega \\ y|_{\partial\Omega} = 0. \end{cases} \tag{4.10}$$

We first note that

Theorem 4.5. *If $y_d \leq 0$ the optimal solution is the function $a = 0$, so the solution is a shape because it is the characteristic function of the empty set, $\chi_\emptyset$.*

Proof. This result is trivial because the function $a \in \overline{\mathcal{U}_L}$ is positive, thus the solution y_a is positive in Ω because $-\Delta$ is a positive operator. If the function $y_d \leq 0$, the minimizer of $\mathcal{J}$ is $y_a = 0$ associated to $a = 0$ because we have a sum of two positive function squared so for sure $y_a - y_d > -y_d$ for every $y_a \neq 0$. □

We have the following results

Theorem 4.6. *If $y_d \geq y^1$ then there exists a unique optimal control $\bar{a}$ that is a shape, $\bar{a} = \chi_\omega$, with $|\omega| = L|\Omega|$.*

Proof. From Theorem 4.4 we know that the optimal control $\bar{a}$ has the form (4.7). We suppose by contradiction that the optimal control $\bar{a}$ is of the form (4.7) with c not zero, i.e

$$\bar{a} = \chi_{\{\bar{p}>s\}} + c\chi_{\{\bar{p}=s\}} \qquad c \neq 0.$$

The solution $y_{\bar{a}}$ associated with $\bar{a}$ is the solution of the system

$$\begin{cases} -\Delta y_{\bar{a}} = \chi_{\{\bar{p}>s\}} + c\chi_{\{\bar{p}=s\}} \quad in\ \Omega \\ y|_{\partial\Omega} = 0. \end{cases} \tag{4.11}$$

We consider the solution y_{a^*} associed to the control $a^* = \chi_{\{\bar{p}>s\}} + \chi_{\{\bar{p}=s\}}$, with $c = 1$, that is

$$\begin{cases} -\Delta y_{a^*} = \chi_{\{\bar{p}>s\}} + \chi_{\{\bar{p}=s\}} \quad in\ \Omega \\ y|_{\partial\Omega} = 0. \end{cases} \tag{4.12}$$

We have

$$-\Delta(y_{\bar{a}} - y_{a^*}) = -\Delta y_{\bar{a}} - (-\Delta y_{a^*}) = \chi_{\{\bar{p}>s\}} + c\chi_{\{\bar{p}=s\}} - \chi_{\{\bar{p}>s\}} - \chi_{\{\bar{p}=s\}} = (c-1)\chi_{\{\bar{p}=s\}} \leq 0.$$

Since $-\Delta(y_{\bar{a}} - y_{a^*}) \leq 0$, by the maximum principle the maximum of $y_{\bar{a}} - y_{a^*}$ is reached on the boundary. For the homogeneous Dirichlet boundary conditions we have that $y_{\bar{a}}$ and y_{a^*} are equal zero at the boundary, and so $y_{\bar{a}} - y_{a^*} \leq 0$ on Ω. So

$$y_{\bar{a}} \leq y_{a^*} \qquad in\ \Omega.$$

For the solution y^1 of (4.10) we have

$$-\Delta(y_{a^*} - y^1) = \chi_{\{\bar{p}>s\}} + \chi_{\{\bar{p}=s\}} - \chi_\Omega = -\chi_{\Omega\setminus(\{\bar{p}>s\}\cup\{\bar{p}=s\})} \leq 0,$$

and for the same reasoning, we deduce that

$$y_{a^*} \leq y^1 \qquad in\ \Omega.$$

If we take $y_d \geq y^1$ then $y_d \geq y_{a^*} \geq y_{\bar{a}}$. And so we have that $(y_{a^*}(x) - y_d(x)) \geq (y_{\bar{a}}(x) - y_d(x))$ for every x.

Since they are two negative numbers for every x we have that $|y_{\bar{a}}(x) - y_d(x))| \geq |(y_{a^*}(x) - y_d(x))|$, and so

$$\frac{1}{2}\left\|y_{a^*} - y_d\right\|^2_{L^2(\Omega)} \leq \frac{1}{2}\left\|y_{\bar{a}} - y_d\right\|^2_{L^2(\Omega)}.$$

There is a contradiction because we supposed $\bar{a}$ the minimum.

So we have that if $y_d \geq y^1$, since the optimal control has to have the form (4.7), the function $\bar{a} = a^*$ is the optimal control and we have that it is a characteristic function, $\bar{a} = \chi_{\{\bar{p}\geq s\}}$.

We call $\omega = \{\bar{p} \geq s\}$ and we prove that $|\omega| = L|\Omega|$. We suppose for the purpose of contradiction that the optimal ω has volume $|\omega| < L|\Omega|$ and we consider a set $\tilde{\omega} \supset \omega$ with $|\tilde{\omega}| = L|\Omega|$. For the maximum principle we have that $y_{\tilde{\omega}} > y_\omega$, thus $y_\omega < y_{\tilde{\omega}} \leq y_d$ and as above we have $\mathcal{J}(\tilde{\omega}) < \mathcal{J}(\omega)$. This is a contradiction because we supposed ω is the optimal set. □

In the following proofs all the equalities are meant to be read as equalities almost everywhere.

Theorem 4.7. *If $y_d \in H^2(\Omega)$ and $-\Delta y_d \leq 0$ or $-\Delta y_d \geq 1$ there exists a unique optimal control $\bar{a}$ that is a shape, $\bar{a} = \chi_\omega$.*

Proof. From the optimality conditions and the Bathtub principle we have that the optimal control $\bar{a}$ has the form:

$$\bar{a} = \chi_{\{\bar{p}>s\}} + c\chi_{\{\bar{p}=s\}}.$$

We recall the second optimality condition from (4.5) is

$$\Delta\bar{p} = y_{\bar{a}} - y_d, \qquad \bar{p} = 0 \ in \ \partial\Omega.$$

We know that in the set $\{\bar{p} = s\}$, $\bar{p}$ is constant so

$$\Delta\bar{p} = 0 \quad in \ \{\bar{p} = s\}.$$

From the second optimality condition $\Delta\bar{p} = y_{\bar{a}} - y_d$ and we have that in the set $\{\bar{p} = s\}$ it must be $y_{\bar{a}} = y_d$.

We have that in $\{\bar{p} = s\}$, it holds that

$$-\Delta y_{\bar{a}} = -\Delta y_d = \bar{a},$$

so $-\Delta y_d \in [0, 1]$ in $\{\bar{p} = s\}$.

If we consider a function y_d such that $-\Delta y_d \geq 1$, we have only two possibility:

1) $-\Delta y_d = 1$ in $\{\bar{p} = s\}$ so the optimal control has to be $\bar{a} = 1$ in $\{\bar{p} = s\}$, and the minimum is a characteristic function

$$\bar{a} = \chi_{\{\bar{p}\geq s\}}.$$

2) $-\Delta y_d > 1$ in $\{\bar{p} = s\}$, since it has to be equal $\bar{a} \in [0, 1]$ the unique possibility is that $|\{\bar{p} = s\}| = 0$, and the minimum is a characteristic function

$$\bar{a} = \chi_{\{\bar{p}>s\}}.$$

If we consider a function y_d such that $-\Delta y_d \leq 0$, by the same reasoning we have that

1) $-\Delta y_d = 0$ in $\{\bar{p} = s\}$ so $\bar{a} = 0$ in $\{\bar{p} = s\}$ and the minimum is $\bar{a} = \chi_{\{\bar{p}>s\}}$.
2) $-\Delta y_d < 0$ in $\{\bar{p} = s\}$ and we must have $|\{\bar{p} = s\}| = 0$, and the minimum is always $\bar{a} = \chi_{\{\bar{p}>s\}}$.

□

We can show an example for the case $-\Delta y_d = 0$, we note that for every constant function $c \in (0,1)$ if we consider $y_d = c$ we have that the minimum exists, because in the set $\{\bar{p} = s\}$ we must have $y_{\bar{a}} = y_d$ and so $y_{\bar{a}} = c$. This implies $-\Delta y_{\bar{a}} = 0$ in $\{\bar{p} = s\}$ so we have $\bar{a} = 0$ in $\{\bar{p} = s\}$.

Theorem 4.8. *If $y_d \in H^2(\Omega)$ and $-\Delta y_d \in (0,1)$ we have that if $|\{\bar{p} = s\}| = 0$ the minimum is a characteristic function, if $|\{\bar{p} = s\}| > 0$ we have relaxation phenomenon.*

Proof. As in the previous proof, we have that

$$\Delta \bar{p} = 0 \quad in\ \{\bar{p} = s\},$$

and that

$$\Delta \bar{p} = y_{\bar{a}} - y_d\ in\ \{\bar{p} = s\}.$$

Thus we have $y_{\bar{a}} = y_d$ in $\{\bar{p} = s\}$, and

$$-\Delta y_{\bar{a}} = -\Delta y_d = \bar{a} \qquad in\ \{\bar{p} = s\}.$$

If $\bar{a}$ is a characteristic function it is never in $(0,1)$, so it never coincides with $-\Delta y_d$. We have only two possibilities:

1) If $|\{\bar{p} = s\}| = 0$ there is no contradiction and we have that the optimal control is of the form
$$\bar{a} = \chi_{\{\bar{p} > s\}}.$$

2) If $|\{\bar{p} = s\}| > 0$, the optimal control is equal to $-\Delta y_d$ and so it has to be $\bar{a} \in (0,1)$ and it is not a characteristic function.

□

We can show an example of relaxation when $-\Delta y_d \in (0,1)$ and also $y^0 < y_d < y^1$. We consider a function y_d such that $-\Delta y_d = f \in (0,1)$ and $y_d|_{\partial\Omega} = 0$. If the function $f \in \overline{\mathcal{U}_L}$, we have that the solution of our relaxed minimization problem is $\bar{a} = f$, because the function $y_{\bar{a}} = y_d$ and $\mathcal{J}(\bar{a}) = 0$. We know that the functional $\mathcal{J}$ is always positive so we have that $\bar{a}$ is the minimizer and it is not a characteristic function. In this case in fact the set $\{\bar{p} = s\} = \Omega$ because the two functions coincide everywhere.

We give here an explicit counterexample in $\mathbb{R}^2$, we consider $\Omega = B(0,1)$ the unit ball in $\mathbb{R}^2$ and

$$y_d = \frac{c}{4}(1 - \|x\|^2),$$

with $0 < c < 1$.

We are in dimension two so $x = (x_1, x_2)$, we compute that

$$\nabla y_d = (-2x_1, -2x_2)$$

and

$$-\Delta y_d = \frac{c}{4}4 = c.$$

We search the minimizer in the set

$$\overline{\mathcal{U}_L} = \{a \in L^\infty(\Omega) \ \mid \ 0 \leq a \leq 1, \ \int_\Omega a(x)dx \leq L|\Omega|\},$$

and if we consider the constant function $a = c$, it belongs to $\overline{\mathcal{U}_L}$ if $c < L$ and we have that

$$-\Delta a = c,$$

exactly as $-\Delta y_d = c$. So we have that $y_a = y_d$ and the functional $\mathcal{J}$ is zero. We have that this is the minimum because the $\inf \mathcal{J}$ is zero and $a = c$ reached zero. The function $a = c$ is not a shape so we have the relaxation phenomenon.

This holds true in higher dimension if we consider the function

$$y_d = \frac{c}{2n}(1 - \|x\|^2),$$

with $0 < c < 1$ and n the dimension of the space.

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